

Pokhara University Faculty of Science and Technology	
Course Code: CVR 250 (3 Credit)	Full Marks: 100
Course Title: Rural Infrastructure Engineering (3-1-0)	Pass Mark: 45
Nature of the Course: Theory	Total Lectures: 45 hours
Level: Bachelor/ Year: II/ Semester: IV	Program: BE Civil & Rural

1. Course Description:
This course is designed to make the students familiar with rural infrastructures and their roles and relevance in rural development. The course covers the functional requirements of rural infrastructures and their planning and design considerations.
2. General Objectives:
The objective of this course is to make the students: - Identify appropriate combination of rural infrastructures supporting the rural development needs in different contexts. - Carry out necessary investigations, analysis, planning and design of trail bridges, rural water supply and sanitation system, rural housing, Farm machinery, post-harvest technology, renewable energy and storage structures.
3. Methods of Instructions:
Lectures, Tutorials, Discussion, Readings and Demonstration works

4. Course Contents	
Specific Objective	Contents
Identify the rural infrastructure in Nepal	Unit 1:Introduction (2 hrs) 1.1 Role of rural infrastructures in rural development 1.2 Development priorities and policies for rural infrastructure development 1.3 Rural infrastructure development programs in Nepal 1.4 Type of rural infrastructure(
Identify the necessity and planning of Rural Energy for rural development	Unit 2:Rural Energy infrastructures (4 hrs) 2.1 Energy Sources and Their Classification 2.2 Solar Energy: Principle 2.3 Solar cooker and crop dryer 2.4 Solar Lantern, Home/Street Lighting, Water Pumping, Refrigeration & Cold Storage 2.3 Biomass Energy Technologies: 2.3.1 Definition and Potential characteristics of Biomass, 2.3.2 Operation and Maintenance of Biogas 2.4 Wind Energy for agricultural application

	and Wind Mill 2.5 Micro hydropower
Understand the planning and design parameters of rural water supply and sanitation system	Unit 3: Rural Water Supply and sanitation (5 hrs) 3.1 Quantity and Quality Consideration in Rural Water Supply 3.2 Intake Works and type of low cost water treatment system; bio sand filter 3.3 Design of gravity flow water supply system with Peak Pressure Tank including Distribution System with Ferro-cement Tank 3.4 Maintenance and upkeep of traditional water supply system 3.5 Rainwater harvesting technology for agriculture purpose and ground water recharge 3.6 Solid Waste Disposal 3.7 Pit Latrine, Ecosan toilets, Septic tanks and Soak Pits.
Describe and Work out planning and design of Rural Housing and Storage structure	Unit 4: Rural Housing and Storage structure (7 hrs) 4.1 Introduction, Purpose and types of farm buildings 4.2 Environmental considerations in the planning of farm buildings and Farmers' residence 4.3 Selection of materials and design considerations for low cost farm buildings 4.4 Environmental considerations for Storage structures for farm inputs- fertilizers, seed, pesticides. 4.5 Rural cold storage system for dairy products and perishables 4.6 Design Parameters for Bag Storage Structures & Bulk Storage Structures

Describe & planning of Farm Mechanization for rural development	Unit 5: Farm Mechanization (8 hrs) 5.1 Primary and Secondary Tillage Implements 5.2 Types and constructional details of recent advances in seeding and planting machines 5.3 Types and constructional details of Potato Digger, Sugarcane Harvester, Fruit Harvesting Machinery 5.4 Types of threshers, and their working principles 5.5 Classification and functional components of Combine harvesters 5.6 Selection criteria of Farm Machines and Equipment for hill and terai Region
Describe & planning of Post-Harvest Technology for rural development	Unit 6: Post Harvest Technology (6 hrs) 6.1 Principle of Cleaning and Grading of Seeds and Grains 6.2 Principle of Separation, Mechanical Separation- equipment and process 6.3 Mixing, Blending and Emulsification, Equipment 6.4 Conveyers- Belt Conveyer, Apron Conveyer, Screw Conveyer, Pneumatic Conveyer 6.5 Moisture Content of Grains and Moisture Measurement Techniques 6.6 Importance of Post-Harvest Technology for rural development
Detail investigations, analysis, planning and design of trail bridges	Unit 7: Trail Bridge (7 hrs) 7.1 General standard for trail bridges 7.2 Classification of trail bridges 7.3 Site investigations for trail bridges 7.4 Foundation of trail bridge and foundation design consideration 7.5 Cable selection criteria 7.6 Selection of Anchor and Tower 7.7 Selection of Clamps and other Accessories 7.8 Trail bridge design practice in Nepal
Planning and design consideration of Gravity rope way	Unit 8: Gravity ropeway (3 hrs) 8.1 Introduction of gravity rope way and its type

	8.2 Site selection and Design consideration 8.3 Maintenance plan and techniques 8.4 Applications in Agricultural and Rural Development
Describe the concept of Rural Road with Green Road Concept	Unit 9: Rural Road (3 hrs) 9.1 Approach to Mule track & Rural Road Planning 9.2 Green Road Concepts 9.3 Context and Basic Principle, Planning, Construction Technology 9.4 Slope protection technique : bio Engineering & SALT technology for road side protection

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester End	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the

requirements defined for the completion of the course.

6. List of Tutorials	
SN	
1.	Numerical on design of Trail Bridge as based on Nepal Standard
2.	Numerical on design of gravity flow rural water supply and sanitation system
3.	Numerical on design consideration of Gravity ropeway
4.	Numerical on design of Storage structures

7. Demonstration	
SN	
1.	Farm Mechanization technology
2.	Post-Harvest Technology
3.	Renewable Energy

8. Prescribed Books and References	
Text Books:	
1. Transportation Engineering by V.N. Vazirani and S.P. Chandola. Khanna Publishers, New Delhi	
2. Elements of Farm Machines, 1 st edition by A.C. Srivastava. Oxford and IBH Publishing Co. Ltd., New Delhi	
3. .M. Mittal. Biomass Systems- Principle and Applications. New Age International Pvt. Ltd., New Delhi	
4. Agricultural Processing Engineering by Handerson, S.M. and R.L. Perry. AVI Publishing Co. Inc., Westport, Connecticut	
References:	
1. Handbook of low cost housing by A.K. Lal. New Age International, New Delhi.	
2. Post-Harvest Technology of Cereals and Pulses by A. Chakravarti. McGraw Hill Inc	
3. H.P. Garg and J. Prakash. Solar Energy Fundamentals and Applications. Tata McGraw Hill Publishing Co., New Delhi (latest edition)	
4. Principle of Farm Machinery, 3 rd edition by R.A. Kepner, Roy Bainer and E.L. Barger. C & S Publishers and Distributors, New Delhi	
5. Trail Bridge Manual Published by DOLI.	
6. Water Supply and Sanitation for Rural and Low Income Urban Communities. Center for Water Supply and Waste Management, BUET, Bangladesh	
7. Thomas and Jordon Jr. Hand Book of Gravity Flow, Surveying, Designing and Construction with Reference to Projects Implemented by Local Development Ministry, Government of Nepal	
8. Suspended Bridge Manual by HELVETAS/Nepal	

Pokhara University Faculty of Science and Technology	
Course Code: GTE 252 (3 Credit)	Full Marks: 100
Course Title: Soil Mechanics (3-2-2)	Pass Mark: 45
Nature of the Course: Theory and Practical	Total Lectures: 45 hours
Level: Bachelor/ Year: II/ Semester: IV	Program: Bachelor of Civil Engineering

1. Course Description:
The course in Soil Mechanics is designed for the bachelor's degree in civil engineering. It is the first series in Geotechnical Engineering course and is a core course for civil engineering in every college/university across the globe. The course covers each and every aspect of soil mechanics starting from the origin of the soil, its phases, properties and behavior under different conditions, its strength to the slope stabilization techniques. The course mainly emphasizes the study of the physical properties of soils and the relevance of these properties as they affect soil strength, stability, drainage, etc.
2. General Objectives:
- This course deals with the fundamental principles governing soil behavior, exploring critical geotechnical properties and various geotechnical applications. - An understanding of these basic concepts enhances the ability of students for the design of foundations of structures, retaining walls, tunnels, excavations, earth fills, stability of earth slopes etc.
3. Methods of Instructions:
Lecture, Tutorial, Discussion, and Laboratory tests.

Specific Objectives	4. Course Contents
At the end of Unit 1, students will be able	Unit 1: Introduction to soil, its Phases, Properties, and Classification (8 hrs)
- To know and explain about rock, soil, rock cycle. - To define the soil and describe its formation process and types. - To visualize the Phase structure of soil and to derive the functional relationships. - To determine the different index properties of soil in laboratory - To know and explain about the types of clay minerals - To classify the soil using various systems based on the properties.	1.1 Origin of Rocks and Soils; Rock cycle 1.2 Definition of soil, Soil formation process, and its types. 1.3 Phase Diagram of Soil: Basic Definitions, Functional Relations 1.4 Index Properties of soil and their determination: Density, Specific gravity and moisture content of soil, Particle shape and size, Grain Size Distribution: Sieve analysis and Hydrometer Analysis, Relative Density of Soil, Consistency Limit of Soil and their Indices, Clay minerals: Montmorillonite, Kaolinite, and Illite, and Soil Classification Systems (Particle size, Textural, USCS, ISCS, AASHTO)
At the end of Unit 2, students will be able	Unit 2: Compaction of Soil (5 hrs)
To define and know about the	2.1 Definition, and purpose of Compaction

definition and purpose of compaction in the field of civil engineering. • To relate dry density with varying moisture content in soil in order to obtain compaction curve and its features. • To conduct laboratory test to determine the compaction characteristics of soil. • To analyze the factors affecting the compaction of soil. • To apply the different methods of compaction based on the site condition. • To determine different methods to determine field density. • To describe the behavior of soil while compacted at dry of optimum and wet of optimum	2.2 Dry density and moisture content relationship 2.3 Laboratory Compaction Tests 2.4 Factors affecting compaction of soil 2.5 Effects of compaction on the engineering behavior of soil 2.6 Methods of Field Compaction 2.7 Determination of field density: Core Cutter method, Sand Cone method, Rubber balloon method, Water displacement method 2.8 Relative Compaction
At the end of Unit 3, students will be able	Unit 3: Permeability, Capillarity, Seepage, and Effective stress, (10 hrs)
• To know the concept of permeability • To determine coefficient of permeability of soil in laboratory and field as well. • To calculate equivalent permeability of stratified soil. • To know the concept of surface tension • To know the concept of capillary rise through soil and to calculate capillary rise. • To analyze the concept of quick sand condition and liquefaction while designing foundation. • To analyze seepage through soil • To analyze failure of an earthen structure through piping mechanism and to apply its preventive measures. • To calculate effective stress in soil and its effects on the strength of soil.	3.1 Permeability, Darcy's law 3.2 Laboratory and field determination of Permeability 3.3 Surface tension and Capillary rise of water in soil 3.4 Quick Sand Condition and Liquefaction 3.5 Equivalent permeability in stratified soil 3.6 Definition of Seepage, and Laplace equation for two-dimensional flow 3.7 Introduction to flow net, their characteristics, applications, and analysis of seepage discharge 3.8 Seepage through Earthen Dam, Concept and construction of Phreatic Line. 3.9 Piping and its Preventive Measures 3.10 Principles of Effective Stress and its effects on soil strength
At the end of Unit 4, students will be able	Unit 4: Stress Distribution on Soil below the Applied Vertical Load (6 hrs)
• To calculate the vertical stress developed at any depth or at any radial	4.1 Concept of Stress Distribution on Soil, Boussinesq's and Westergaard's Theory

distances due to the load applied on the ground surface using various stress distribution theories and related charts/tables. - To draw and know the variation of the vertical stress distribution on horizontal and vertical plane below the ground surface. - To locate the significant depth of foundation using the concept of pressure bulb.	4.2 Concept of Pressure Bulb and Vertical Stress Distribution on horizontal & vertical plane 4.3 Vertical Stress Distribution beneath the loaded area (Line load, Strip load, circular area, and Rectangular area), Newmarks and Fadum's chart 4.4 Approximate Method of Stress Distribution
At the end of Unit 5, students will be able	Unit 5: Compressibility and Consolidation of Soil (8 hrs)
- To define and understand the principles of consolidation, its type, and related theories. - To know the mechanism of consolidation given by Terzaghi's spring analogy model. - To calculate the time of consolidation - To conduct consolidation test in the laboratory. - To draw void ratio pressure relationship - To determine consolidation parameters of saturated soil - To know the types of consolidated clay and to determine pre-consolidation pressure - To determine consolidation settlement	5.1 Definition, Principles of consolidation, its type 5.2 Terzaghi's spring analogy model 5.3 One-Dimensional Consolidation theory 5.4 Consolidation test 5.5 Void ratio pressure relationships: Compressibility Characteristics (i.e. Compression Index, Coefficient of Compressibility, Recompression Index, Swelling Index, Coefficient of Volume change, etc.) 5.6 Normally Consolidated Clay and Over Consolidated Clay and determination of Over Consolidation (i.e. Pre-consolidation) Pressure 5.7 Determination of Consolidation Settlement 5.8 Determination of Coefficient of Consolidation by square root and logarithm of time fitting method
At the end of Unit 6, students will be able to understand about	Unit 6: Shear Strength Soil, and Slope Stabilization Techniques (8 hrs)
- To know the shear strength of soil and governing parameters. - To understand the Mohr-Coulomb's theory, plane of failure and principle planes and stresses. - To determination the shear strength parameters conducting different laboratory test. - To determine shear strength of clay and sand. - To know the types of slope and to find the causes and mechanism of slope failure.	6.1 Concept of shear strength of soil 6.2 Mohr-Coulomb's Failure Criteria 6.3 Mohr Circle, Normal Stress, Shear Stress, Principle planes, Principle stresses, and plane of failure 6.4 Determination of Shear Strength Parameters: (i) Direct Shear Test, (ii) Triaxial test: Consolidated Drained (CD) Test, Consolidated Undrained (CU) test, Unconsolidated Undrained (UU) Test, (iii) Unconfined Compression Test, and (iv) Vane Shear Test 6.5 Shear Strength of Clay and Sand 6.6 Types of slope, cause and mechanism of slope failure

- To determine the FOS and critical surface. - To stabilize the slopes using different techniques.	6.7 Factor of safety and critical surface 6.8 Slope Stabilization Techniques
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5. List of Tutorials	
SN	Numerical problems related to
1.	Phase relationship and properties of soil
2.	Soil classification
3.	Compaction characteristics
4.	Effective stresses on hydrostatic, uniform seepage, Capillary, and uniform surcharge condition
5.	Determination of permeability of soil
6.	Computation of pore pressure at any point of flow net and Calculation of Seepage discharge through soil
7.	Computation of vertical stresses due to applied load (point load, line load, strip load, and UDL on circular Rectangular area)
8.	Use of Newmark and Fadum's chart to compute vertical stress
9.	Consolidation Settlement and Time Rate of Consolidation
10.	Shear strength of soil (c , ϕ , τ , principal planes, principal stresses, etc.)

6. List of Practical	
SN	List of Practical
1.	Determination of specific gravity and water content of soil
2.	Grain size analysis of soil (i.e. Sieve Analysis and Hydrometer Analysis)
3.	Determination of wet density of soil.
4.	Consistency Limits Test for fine-grained soil (i.e. LL, PL, and SL)
5.	Compaction Test
6.	Direct Shear Test
7.	Triaxial Test (Demonstration)
8.	Unconfined Compression Test
9.	Vane Shear Test
10.	Consolidation Test (Demonstration)
11.	Visual Classification Test of Soil (Demonstration)
12.	Experimental demonstration of water effect on sandcastle

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester End	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References
Name of Authors / Name of Book / Publishers
Text Books: 1. Murthy, V.N.S. (2007). <i>Text Book of Soil Mechanics and Foundation Engineering (Geotechnical Engineering Series)</i> , CBS Publishers and Distributors Pvt. Ltd. India 2. Ranjan, Gopal & Rao, A.S.R. (2000), <i>Basic and Applied Soil Mechanics</i> , New Age International Publishers, New Delhi, India.

References:

1. Terzaghi, Karl, Peck, R.B. & John, Wiley (1967). *Soil Mechanics in Engineering Practice*, New York.
2. T. William Lambe, Robert V. Whitman (1969), *Soil Mechanics*, JOHN WILEY & SONS, New York
3. T. William Lambe, *SOIL TESTING for Engineers*, John Wiley & Sons, Inc. New York
4. Dante Fratta, Jennifer Aguetant, Lynne Roussel-Smith (2007), *Introduction to Soil Mechanics Laboratory Testing*, CRC Press, New York
3. Braja M. Das (2002), *Principles of Geotechnical Engineering*, Thomson, Asia
4. Venkatramaiah, C. (Third Edition), *Geotechnical Engineering*, New Age International (P) Limited Publisher, India
5. Punmia, B.C, Jain, A.K.& Jain, Arun K. (Seventh Edition 2017). *Soil Mechanics and Foundation engineering*, Laxmi Publication Pvt. Ltd. India.
6. Adhikari P.B., Thapa H.J. A Text Book of Soil Mechanics, (2023), Sanskriti Prakashan, Kathmandu, Nepal

Pokhara University Faculty of Science and Technology	
Course Code: MGT 250 (3 Credit)	Full Marks: 100
Course Title: Engineering Economics (3-1-0)	Pass Mark: 45
Nature of the Course: Theory and Tutorial	Total Lectures: 45 hours
Level: Bachelor/ Year: III/ Semester: VI	Program: Bachelor in Civil Engineering

1. Course Description:

The course introduces concepts and economic analysis procedures to assist with decision-making in engineering projects. Concepts include fundamentals of economics and engineering economics, cost analysis, time value of money and cash flow diagrams; interest rates, different techniques of evaluation, risk, replacement, depreciation and inflation. Economic analysis procedure includes selection and ranking of projects and analyze the associated risk including, replacement analysis, calculation of depreciation amount and adjustment of taxes and inflation.

2. General Objectives:

The basic purpose of this course is to provide a sound understanding of concepts and principles of economy and engineering economy and to develop proficiency with methods for making rational decisions regarding financial analysis of engineering problems likely to be encountered in professional practice.

3. Methods of Instructions:

Lecture, tutorials, discussions, assignments, quizzes, project work

Specific Objective	4 Course Content
- Explain the meaning and principles of economics - Describe the law of demand, Law of supply, production and utility theory - Explain the meaning and importance of engineering economics and its principles	Unit 1: Basics of Engineering Economics (3 hrs)
	1.1 Definition of economics and principles of economics
	1.2 Introduction to demand, supply, production, and utility
	1.3 Definition of engineering economics, principles of engineering economics and its applications.
Describe various elements and types of cost	1.4 Terminologies used in engineering economic analysis
	Unit 2: Cost Concept and Analysis (3 hrs)
	2.1 Elements of cost-material, labor and expenses
	2.2 Various types of costs: Direct cost, indirect cost, fixed cost, variable cost, marginal cost, opportunity cost, sunk cost, manufacturing cost and non manufacturing cost.

	Unit 3: Interest and Time Value of Money (6 hrs)
- Define different types of interest rates and describe their relationship. - Develop interest formulas for equivalent calculations of different types of cashflows	3.1 Simple interest, compound interest, nominal interest rate and effective interest rates 3.2 Economic equivalence: Definitions, simple calculations, and general principles 3.3 Development of formulas for equivalence calculations: Types of cashflows, single cashflow formulas, equal payment series, linear gradient series, geometric gradient series, irregular (mixed payment) series.
	Unit 4: Basic Methodologies of Engineering Economic Studies (6hrs)
- Define and describe minimum attractive rates of return - Explain the process to calculate Payback period, equivalent worth, internal and external rates of return as well as Benefit cost ratio and their interpretation - To explain the meaning and differences of economic and financial analysis	4.1 Minimum attractive rate of return (MARR) : Concept and its calculation 4.2 Payback period: Simple and discounted 4.3 Equivalent worth: Present worth, future worth and annual worth 4.4 Rate of return: Internal rate of return (IRR) and External rate of return (ERR) 4.5 Benefit cost ratio: Normal, conventional and modified 4.6 Economic and financial analysis: Basic concept
	Unit 5: Comparative Analysis of Alternatives (6 hrs)
- To be able to rank and select the projects having same and different useful lives economically and financially based on values obtained from Payback period, Equivalent worth, rate of return and benefit cost ratio - To be able to select combinations of projects.	5.1 Concept of mutually exclusive, independent and contingent projects 5.2 Comparing alternatives having same useful life 5.2.1 Payback period, present worth and future worth methods 5.2.2 IRR, ERR and BCR method using incremental analysis 5.3 Comparing alternatives having different useful lives: 5.3.1 Repeatability assumptions 5.3.2 Co-terminated assumptions 5.3.3 Capitalized worth method 5.4 Selecting best project for different combinations of project among mutually exclusive, independent and contingent project
	Unit 6: Replacement Analysis (4 hrs)
- To explain the process to calculate the economic service life of the machine - To be able to decide the right time to replace the existing	6.1 Basic concept of defender and Challenger 6.2 Economic service life 6.3 Replacement strategies under finite and infinite planning horizon

machine by new one.	
	Unit 7: Risk Analysis (6 hrs)
- To define risk and their origin in Engineering projects - To explain different methods of dealing with project risk and their application	7.1 Concept and sources of risk in engineering project 7.2 Methods of dealing with risk: 7.2.1 Breakeven analysis 7.2.2 Sensitivity analysis 7.2.2 Decision tree analysis 7.2.4 Scenario analysis 7.2.5 Concept of Monte Carlo simulations
	Unit 8: Depreciation and Taxes (6 hrs)
- Define depreciation, depletion, their causes and explain their application - Explain the process to calculate depreciation and corresponding Book values uses various methods. - Define different types of taxes - Explain the process to prepare after tax cashflow and profitability	8.1 Depreciation and depletion, causes and application 8.2 Methods of depreciation: Straight line method; Units of production method; Sum of year digit method; Declining balance methods; MACRS method and Sinking fund method 8.3 Depletion: Cost depletion and percentage depletion 8.4 Taxes: Direct and indirect tax, tax rates and VAT 8.5 After tax cashflow calculation, NPV and IRR calculations
	Unit 9: Capital Budgeting Decision (3 hrs)
- Define capital budgeting decision - Explain sources of fund/method of financing with their advantages and disadvantages - To calculate the cost of capital	9.1 Define capital budgeting their importance 9.2 Methods of financing: Equity financing, debt financing and Capital structure their merits and demerits 9.3 Cost of capital: Cost of equity, cost of debt and cost of capital
	Unit 10: Inflation (2 hrs)
- Define inflation, their causes and explain the methods to explain inflation - To derive the relationship between inflation, Interest rate and inflation adjusted interest rate - Define constant dollar and actual dollar, explain their relationship	10.1 Meaning and measure of Inflation 10.2 Actual vs constant dollar 10.3 Market and inflation free interest rates 10.4 Constant dollar and actual dollar analysis

5. List of Tutorials	
SN	Tutorials
1.	Application of simple interest, compound interest, nominal interest rate and effective interest rates; Calculation of present value, future value, annual value for different types of

	cashflows (Unit 3)
2.	Payback period (simple and discounted), Net present value, Net Future Value, Net Annual Value, IRR/ERR, Benefit Cost Ratio (Conventional and modified) calculation (Unit 4)
3.	Ranking of project using all methods and perform incremental analysis for projects having same useful life where applicable; Ranking of projects having different useful lives (Repeatability, co- terminated and capitalized worth methods); Selection of combinations of projects (Unit 5)
4.	Find Economic service life, Finding best replacement strategy for finite and infinite planning horizon. (Unit 6)
5.	Breakeven analysis, Sensitivity analysis and Decision tree analysis (Unit 7)
6.	Calculate depreciation amount and corresponding book values using different methods of depreciation; Prepare after tax cash flow and find corresponding NPV and IRR. (Unit 8)
7.	Calculate cost of capital for debt and equity financing (Excluding and including tax) (Unit 9)
8.	Calculate average inflation, Relationship between inflation, interest rate and inflation free interest rates; Find NPV and IRR for Actual dollar and Constant dollar cashflow (Unit 9)

6. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of attendance/class participation, assignment, project report and internal Assessment. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		50	Semester End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Project Report	10%			
Internal Assessment	60%			
Total Internal		50		50
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books:

1. Fundamentals of Engineering Economics by Chan S Park; Pearson
2. Engineering Economy by by [William Sullivan](#) , [Elin Wicks](#) , [C Koelling](#); Pearson

References:

1. A text book of Engineering Economics by OP Giri
2. Principles of Engineering Economic analysis by D Adhikari,
3. Fundamentals of Engineering Economic Analysis by SK Shrestha, Bikash Gautam

Pokhara University
Faculty of Science and Technology

Course code: MTH 250 (3 Credits)
Course title: Applied Mathematics (3-2-0)
Nature of the course: Theory
Level: Bachelor

Full marks: 100
Pass marks: 45
Total lectures: 45 hrs.
Program: BE

1. Course Description

This course is designed for developing competency of the students in the applications of various mathematical concepts they learned in courses in previous semesters. It is equipped with complex analysis, Z-transform, Partial differential Equations and Fourier Transform. The pre requisite for this course is Calculus I, II and Algebra and Geometry. The course will be delivered through lecture method, assignment on practically base engineering problems and class tests.

2. General Objectives

The course is designed with the general objective:

- To acquaint the students with applications of mathematical tools in engineering.

3. Methods of Instruction

Lecture, tutorials, discussions and assignments

4. Contents in Detail

Specific objectives	Contents
• Understand and apply function of complex variables, Calculus of functions of complex variables and their applications in Engineering problems.	Unit I: Complex Analysis (17 hrs.) 1.1 Complex numbers and functions (5 hrs.) 1.1.1 Review on Complex number, their geometric representation, Polar form, power and roots. 1.1.2 Sets and functions in complex plane, Limits Continuity and derivatives of function of complex variables. (Definition and concepts only) 1.1.3 Analytic functions, Cauchy-Riemann(C-R) equations as necessary conditions for functions to be analytic, C-R equations as sufficient condition for analyticity (without proof), Polar form of C-R equations (No derivation). 1.1.4 Laplace equation, harmonic functions and harmonic conjugate 1.1.5 Related problems
	1.2 Integrals in complex plane (4 hrs.) 1.2.1 Line integrals in the complex plane, Evaluation of basic line integrals in complex plane 1.2.2 Cauchy's Integral theorem, Cauchy's integral formula

	and Cauchy integral formula of higher order (for analytic functions) without proof. 1.2.3 Related problems. 1.3 Taylor and Laurent series for functions of complex variables. (6 hrs.) 1.3.1 Taylor series and Laurent series (Without Proof) and Related Problems 1.3.2 Singularities and zeros, Residues and integration, Cauchy Residue theorem (Without proof) and related Problems. 1.4 Conformal mapping (2 hrs.): Special Linear fractional transformation (Bilinear fractional transformation) only.
Understand and apply discrete transforms and solve difference equations.	Unit II: Z-Transform and its Applications (10 hrs.) 2.1 Z-transform, Z-transform of elementary functions, Properties of Z-transforms, Shifting theorems, initial value theorem, final value theorem. 2.2 Inverse z-transforms using division method, expansion method, Partial fraction method and residue method. 2.3 Application: Difference equations and solution by using Z-transform.
Understand and apply higher dimensional systems and describe them by partial differential equations with solution techniques and interpretation of solutions.	Unit III: Partial Differential Equations (12 hrs.) 3.1 Partial differential equations and solutions by variable separation method. 3.2 One dimensional wave equation and its solutions and related problems. 3.3 One dimensional heat equation and its solutions and related problems. 3.4 Two dimensional heat equation, Laplace equation (steady state heat equation) and its solution for rectangular boundaries. Laplace equation in polar form and its solution for circular boundaries, related problems.
Evaluate Fourier integrals and Transforms.	Unit IV Fourier integral and Transform (6 hrs.) 4.1 Fourier integral, Fourier sine and cosine integrals and related problems. 4.2 Fourier integral in complex form and Fourier transform and inverse transform, Fourier sine and cosine transforms and their inverse transforms, Convolution theorem, Parseval's identity and related problems.

Note: The figures in the parentheses indicate the approximate periods for the respective units.

5. List of Tutorials

Tutorial work covers the work to be done in tutorial. This will enable the students to compute the mathematics problem under the supervision of the course leader. The major tutorial works are as follows:

Total : 30 Hours

Unit no.	Unit name	List of Tutorials	Tutorial hours
1	Unit I: Complex Analysis (9 hrs.)	1.1 Problems on differentiability 1.2 Problems on analyticity 1.3 Problems on Harmonic and conjugate harmonic functions. 1.4 Problems on Integrals using Cauchy integral theorem and formula. 1.5 Problems on Taylor's series and Laurent's series 1.6 Problems on singularities and residues.	1 hr. 1 hr. 1 hr. 2 hr. 2 hr. 2 hr.
2	Unit II: Z-Transform and its Applications (7 hrs.)	2.1 Problems on Z-transforms of elementary functions. 2.2 Problems on Z-transforms using different theorems. 2.3 Problems on inverse z-transforms. 2.4 Solution of difference equations.	1 hr. 2 hrs. 2 hrs. 2 hrs.
3	Unit III: Partial Differential Equations (10 hrs.)	3.1 Problems on separation of variables methods. 3.2 Problems related to one dimensional wave equation. 3.3 Problems on one dimensional heat equation. 3.4 Problems on two-dimensional heat equation rectangular boundaries 3.5 Problems on two-dimensional heat equation circular boundaries.	2 hrs. 2 hrs. 2 hrs. 2 hrs. 2 hrs.
4	Unit IV Fourier integral and Transform (4 hrs.)	4.1 Problems on Fourier integrals. 4.2 Problems on Fourier Transforms and its inverse.	2 hrs. 2 hrs.

6. Evaluation System and Students' Responsibilities

Evaluation System

Internal evaluation is done as follows:

Internal Evaluation	Marks	External Evaluation	Weight	Marks
Attendance & Class Participation	10%	Semester End Board Examination	50%	50
Assignments	20%			
Presentations/Quizzes	10%			
Term exam	60%			
Total Internal	50			
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References**Text Book**

1. Advanced Engineering Mathematics, Erwin Kreszig
2. Text Book of Engineering Mathematics, Debashis Dutta , NEW AGE International Publisher

References

1. Advanced Engineering Mathematics, Alan Jeffrey
2. Engineering Mathematics, S.S sastry Vol.1 and Vol.2

Pokhara University
Faculty of Science and Technology

Course Code: STR 252 (3 Credits)
Course title: Structural Analysis I (3-2-1)
Nature of the Course: Theory & Practical
Level: Bachelor

Full Marks: 100
Pass Marks: 45
Total Lectures: 45 hours
Program: BE

1. Course Description

This course provides the basic concept and knowledge of structural analysis of statically determinate structure. It enables students to analyze statically determinate beam, frame, truss, arch and cable both by manual calculation as well as computer simulation.

2. General Objectives

The general objectives of this course are:

- To equip the students with the fundamental concept structural analysis and its practical application.
- To familiarize the students with strain energy, virtual work, slope, deflection, influence line diagram, arch, cable and space truss.

3. Methods of Instruction

Lecture, Tutorial, Discussion, Readings and Practical works

4. Contents in Detail

Specific Objectives	Contents
Know the types of structures, linearity and non-linearity of structural analysis, degree of static and kinematic indeterminacy, understand the role of structural analysis in structural engineering.	Unit I: Introduction (3 hrs) 1.1 Introduction to structural analysis 1.2 Role of structural analysis in structural engineering projects 1.3 Types of structures and structural elements 1.4 Stability and determinacy of structures 1.5 Approaches of structural analysis 1.6 Linearity and non-linearity in structural analysis 1.7 Degree of static and kinematic indeterminacy
Determine the deflection of beams, frames and truss of the structure by work-energy method.	Unit II: Deflection of Beams, Frames and Trusses: Work-Energy Methods (12 hrs) 2.1 Strain energy and complementary strain energy; work and complementary work 2.2 Strain energy due to gradually, suddenly applied direct load: dynamic multipliers 2.3 Strain energy due to axial force, shear force, bending moment and torsion 2.4 Displacement of beam, frame and truss by strain energy method 2.5 Principle of virtual work

	2.6 Deflection of beams, frames and trusses by virtual work method 2.7 Betti's law and Maxwell's law of reciprocal deflections 2.8 Castigliano's theorems and application for beams and plane frames
Determine the slope and deflection of beams by various geometric methods.	Unit III: Slope and Deflections of Beams: Geometric Methods (6 hrs.) 3.1 Double integration method 3.2 Macaulay's method 3.3 Superposition method 3.4 Conjugate beam method 3.5 Moment area method 3.6 Conjugate-beam method
Understand the importance of influence line diagram for analysis of structure, determine reaction, shear force, bending moment at particular point and absolute maximum response of structure by influence line method	Unit IV: Influence Lines for Simple Structures (9 hrs.) 4.1 Influence lines for statically determinate beams 4.2 Influence lines for statically determinate trusses 4.3 Influence lines for girders with floor systems 4.4 Response at a particular location due to a single moving concentrated load 4.5 Response at a particular location due to a uniformly distributed live load 4.6 Response at a particular location due to a series of moving concentrated loads 4.7 Absolute maximum response
Understand the uses of the statically determinate circular and parabolic arch. Solve problems related to electrostatics statically determinate arches by analytical and influence line diagram method.	Unit V: Statically Determinate Arches (6 hrs.) 5.1 Types of arches 5.2 Three-hinged arches with support at same and different level 5.3 Determination of support reactions, normal thrust, radial shear and bending moment of circular and parabolic arches 5.4 Axial force, shear force and bending moment diagrams in three hinged parabolic arches 5.5 Influence line diagrams for reactions, bending moments, radial shear, normal thrust 5.6 Maximum internal forces (axial force, shear force and bending moment) in three hinged parabolic arches
Understand the importance of cable structures, analysis of parabolic cable structures and three-hinged stiffening girder Determine shear force and bending moment of three-hinged stiffening girder by influence line diagram	Unit VI: Cable Structures (6 hrs.) 6.1 Introduction to cable structures 6.2 Elements of a simple suspension bridges 6.3 Analysis of parabolic cables 6.4 Analysis of three-hinged stiffening girder 6.5 Influence line diagrams and determination of shear forces and bending moments for three-hinged

approach.	stiffening girder
Know the practical application of statically determinate space truss Determine the member force of statically determinate space truss	Unit VII: Statically Determinate Space Trusses (3 hrs.) 7.1 Introduction to simple space trusses 7.2 Types of supports 7.3 Determinacy and stability 7.4 Analysis of space truss by tension coefficient method

Note: The figures in the parentheses indicate the approximate periods for the respective units.

5. List of Tutorials

The following tutorial activities of 2 hours/ week per group of maximum 24 students should be conducted to cover all the required contents of this course.

S.N.	Tutorials
1	Determination of static and kinematic indeterminacy of various structures.
2	Solving and analyzing the problems related to work-energy method.
3	Determination of slope and deflection of different types of beam by various geometric methods.
4	Determination of angle of acceptance for working of optical fiber and finding population of atoms in different energy states. Solving the problems of beam and truss structures by influence line diagram method.
5	Solving the problems of three hinged parabolic and circular arch both analytical and influence line diagram method
6	Solving the problems related to suspension cable bridge
7	Determination of support reactions and member forces of space truss.

6. Practical Works (Any Eight)

S.N.	Practical works
1	To determine the deflection of beam experimentally and verify the result by computer simulation.
2	To determine the displacement of frame experimentally and verify the result by computer simulation.
3	To determine the displacement of truss experimentally and verify the result by computer simulation.
4	To measure the effect of Influence lines for beam experimentally and verify the result by computer simulation.
5	To determine the horizontal thrust of three-hinged arch experimentally and verify the result by computer simulation.
6	To determine the tension in cable of suspension bridge by experimentally and verify the result by computer simulation.
7	To determine the support reactions and member forces of space truss by computer simulation.

7. Evaluation system and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student's Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. Kassimali, A. (2009). *Structural Analysis*. Cengage Learning.
2. Norris, C. H., & Wilbur, J. B. (1960). *Elementary Structural Analysis*. McGraw-Hill.

References

1. Bhavikatti, S. S. (2011). *Structural Analysis I*. New Delhi: Vikas Publishing House Pvt. Ltd.
2. Darkov, A. & Kuznetsov, V. (2011). *Structural Mechanics*, Moscow: Mir Publishers.
3. Hibbeler, R.C. (2009). *Structural Analysis*. Pearson.
4. Jain, A.K. (2012). *Strength of Materials and Structural Analysis (2012)*. Roorkee: Nem Chand & Bros.
5. Reddy, C.S. (1999). *Basic Structural Analysis*. Tata McGraw-Hill Education.

Pokhara University Faculty of Science and Technology	
Course Code: WRE 211 (3 Credit)	Full Marks: 100
Course Title: Hydraulics (3 – 2 – 2)	Pass Mark: 45
Nature of the Course: Theory and Practice	Total Lectures: 45 hours
Level: Bachelor/ Year: II/ Semester: IV	Program: Bachelor in Civil Engineering

1. Course Description:

This course deals with the study of hydraulic analysis and its application to hydraulic structures (e.g., pipe network, power turbine, reservoir, dam, spillway, weir etc.) for designing water supply system, hydropower generation, irrigation channels, flood control and other water-related infrastructures. Specifically, this course discusses: different flow phenomena; governing laws, applications for closed conduit flow, and open channel flow systems.

2. General Objectives:

Overall objective of this course is to enable students to analyze flow characteristics in pipe flow as well as in open channel flow systems, which aims to impart the concept of hydraulic phenomena in water resources engineering and their application in the field of civil engineering.

3. Methods of Instructions:

Lecture, Tutorial, Discussion, Readings and Practical works

4. Course Contents

Specific Objective	Contents
- Understanding concept of laminar and turbulent flow in pipes - Enable to understand and derive relationships for shear and velocity distributions in pipe flow - Enable to derive equations for estimation of major (frictional) loss in pipe flow	Unit 1: Laminar and Turbulent Flow in Pipes (6 hrs) 1.1 Introduction to pipe flow, Reynolds experiment and flow based on Reynolds' number 1.2 Laminar flow: Steady-uniform-incompressible flow in a circular pipe; Shear stress and velocity distribution; Loss of head due to friction (Hagen-Poiseuille equation) 1.3 Turbulent flow: Shear stress development; Prandtl's mixing length theory; Velocity distribution; Loss of head due to friction (Darcy- Weisbach equation) 1.4 Hydrodynamically smooth and rough boundaries; Nikuradse's experiment, Variation of friction factor with Reynolds number; Resistance for commercial pipes; Colebrook-White equation; Use of Moody's

	diagram
• Understanding and calculation of various minor losses associated with pipe flow • Understanding of simple and complex pipe flow networks; and their comparative advantages and disadvantages • Understanding and calculation of various flow variables in simple and complex pipe network systems	Unit 2: Pipe Flow Systems (9 hrs) 2.1 Introduction to pipe system (fittings, bends, valves); Minor head losses in pipes (losses in sudden enlargement, sudden contraction, exit loss, entry loss, losses due to sudden obstruction, losses in bends and losses due to different fittings); HGL and TEL lines 2.2 Three types of pipe flow problems and their solution 2.3 Pipe Line System (Pipes in series and parallel): Dupuit's equation, concept of equivalent pipe length/diameter in series and parallel; Concept of economic diameter of pipes 2.4 Siphons: Definition, application, conditions for continuous supply, different type of problem in siphon (simple and trial & error) 2.5 Pipe network solution by Hardy-Cross method for single and double loops of pipe networks
• Understanding concept of unsteady flow and water hammer phenomena, • Enable to compute the rise in pressure due to water hammer, and • Enable to understand the necessity and working of a surge tank	Unit 3: Water Hammer in Pipes (4 hrs) 3.1 Basic concept of unsteady flow; Water hammer, its causes & effects in pipes 3.2 Velocity of pressure wave in a rigid pipe; Propagation of pressure wave; pressure variation with time at different sections 3.3 Water Hammer due to gradual and sudden closure of valve for the cases of rigid and elastic pipes; Equations of pressure rise for water hammer
• Understand how flow in open channels differs from pressurized flow in pipes • Learn the different flow regimes in open channels and their characteristics	Unit 4: Open Channel Flow (3 hrs) 4.1 Introduction; Difference between open channel and pipe flows 4.2 Types of open channel flows: steady & unsteady; uniform & non-uniform flows (gradually, rapidly and spatially varied flows); Sub-critical, critical and super critical flows 4.3 Classification of open channels (natural and artificial channel, prismatic and

	non-prismatic channel, rigid boundary and mobile boundary channel); Geometric properties of open channels (depth, flow area, top width, wetted perimeter, hydraulic radius, hydraulic mean depth, bed slope, hydraulic or energy slope, water surface slope), Shapes of open channel
- Describe concept and conditions of uniform flow and conditions in open channel, - Explain equations for handling of uniflow problems in open channel - Explain conditions and formulae for designing efficient channel sections	Unit 5: Uniform Flow in Open Channels (6 hrs) 5.1 Conditions of uniform flow in a prismatic channel, expression for shear stress on boundary of channel, velocity and shear stress distribution in open channel and mean velocity 5.2 Fundamental equations of uniform flow: Manning's equation and Chezy's equation, relationship between Chezy's coefficients (C), Manning's and Darcy's-Weisbach co-efficient 5.3 Factors affecting manning's roughness coefficient. 5.4 Conveyance, section factor and hydraulic exponent for uniform flow computation 5.5 Determination of normal depth, velocity and slope 5.6 Design of economic channel sections (rectangular, triangular, trapezoidal and circular)
- Understanding concept and derivations of energy and momentum principles in open channel - Explain concept and formulae of specific energy; specific force; conditions for critical flow, maximum discharge - Explain various applications including discharge measurement	Unit 6: Energy and Momentum Principles in Open Channel (6hrs) 6.1 Specific energy, specific energy diagram, critical depth of flow 6.2 Critical depth computations for all kind of channel sections (prismatic) and criteria for critical state of flow 6.3 Alternate depth, depth-discharge relationship 6.4 Application of energy principle and critical depth concept: channel width reduction, rise in channel bed, venture flume and broad crested weir 6.5 Momentum principle, specific force, specific force curve, criteria for critical state

	of flow, conjugate depth
• Explain concept and governing equations of gradually varied flow • Understanding various types of channel bed slopes and flow profiles • Enable to compute gradually varying water surface profiles	Unit 7: Gradually Varied Flows (GVF) and its Analysis (6 hrs) 7.1 Introduction to GVF, reasons and examples of GVF 7.2 Basic assumptions, governing /dynamic equation and its physical meaning 7.3 Classification of channel bed slopes (mild, critical, steep, horizontal and adverse) and Characteristics of flow profiles in prismatic channels 7.4 Computation of GVF in prismatic channels by graphical integration, direct step and standard step methods
• To study the development of rapidly varied flow and hydraulic jump • Enable to investigate the rapidly varied flow in open channel and compute various hydraulic jump variables	Unit 8: Hydraulic Jump and its Analysis (5 hrs) 8.1 Characteristics of Rapidly Varied Flow (RVF) 8.2 Hydraulic jump and its uses as an energy dissipater: jumps in a horizontal rectangular channel, jump variables (conjugate depth, height of jump, length of jump) 8.3 Energy loss in jump 8.4 Classification of the jump based on the tail water level and Froude number

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester End	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			

Viva	30%		
Total Internal		50	
Full Marks: 50 + 50 = 100			

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. List of Tutorials	
SN	
1.	Computation of shear stress, velocity, pressure, flow rate, major and minor head losses for laminar and turbulent flow in pipes
2.	Computation of head loss, flow (Q) and size (diameter) in a simple pipe
3.	Computation of discharge, head loss, pressure in a siphon
4.	Computation of discharge, length, diameter in pipe in series and parallel
5.	Computation of head losses and discharge in pipe network using Hardy-cross method
6.	Computation of discharge, head losses, elevation in case of three interconnected reservoirs
7.	Computation of risk for pipe burst; rise in pressure in pipe for gradual and sudden closure of valves, time of closure of valves
8.	Computation of flow rate, shear stress, velocity, normal depth, slope in open channels
9.	Computation of most economical cross-sections for triangular, rectangular, trapezoidal and circular open channels
10.	Computation of Froude number, normal slope, critical slope, flow rate, alternate depths, specific energy, specific force, critical velocity, critical depths, conjugate depths for flow in open channels
11.	Computation of depth of flow, width reductions, floor rise (height of hump) for critical flow conditions in open channel
12.	Computation of type of flow profiles; characteristics (e.g., depth, distance) for GVF by direct and standard step methods
13.	Computation of jump location, heights, surface profiles for hydraulic jump

7. List of Practicals	
SN	
1.	Reynolds' experiment
2.	Head loss in a pipeline
3.	Flow through open sluice gate
4.	Hydraulic jump analysis in open channel
5.	Hump and constricted flow analysis: discharge measurement in open channel (channel

	width reduction, rise in channel bed and venture flume)
6.	Introduction to open-source hydraulic software (e.g., HEC-RAS)

8. Prescribed Books and References

Text Books:

1. Modi, P. N. & Seth, S. M. Fluid Mechanics and Hydraulics. New Delhi: Standard Books.
2. Subramanya, K. Flow in Open Channel. New Delhi: Tata McGraw Hill.
3. Bansal, R. K. A text book of Fluid Mechanics and Hydraulic Machines, New Delhi: Laxmi Publications.

References:

1. Chow, V.T. Open Channel Hydraulics, New Delhi: McGraw-Hill.
2. K. G. Ranga Raju. Flow through Open Channel. New Delhi: Tata McGraw Hill Publishing Company Ltd.
3. Jain, A. K. Fluid Mechanics and Hydraulics. New Delhi: Khanna Publication.
4. Kumar, D.S. Fluid Mechanics and Fluid Power Engineering. Delhi: S.K. Kataria and Sons.
5. Rajput, R. K. Fluid Mechanics and Hydraulic Machines. New Delhi: S. Chand.
6. Sangraula, D. P. & Bhattarai, P. A text book of Hydraulics.

Pokhara University
Faculty of Science and Technology(Revised)

Course No.: CMP 272

Full marks: 100

Course title: Analysis and Design of Algorithms **(3-1-3)**

Pass marks: 45

Nature of the course: Theory & Practical

Total Lectures: 45 hrs

Level: Bachelor

Program: BE (Software)

1. Course Description

This course is designed to provide the students with the knowledge of how the algorithms are designed and how their performances are analyzed. This course focuses on the analysis of algorithms using asymptotic analysis techniques for performance analysis and various algorithm design techniques such as divide and conquer, dynamic programming, greedy approaches and backtracking techniques.

2. General Objectives

- To acquaint the student with the knowledge of the basic concept of algorithmic design techniques that are used to solve real world problems.
- To familiarize the students with the concept of computational complexity and algorithm analysis techniques.
- To develop the skill in students to design new algorithms and to analyze them to choose a more efficient algorithm.

3. Methods of Instruction

Lectures, Tutorials, Case Studies, Discussion, Readings and Practical Works.

4. Contents in Detail.

Specific Objectives	Contents
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● Use the asymptotic notation for algorithm analysis	Unit 1: Introduction (6 hours) 1. Introduction to Algorithm 2. Algorithm Specification ● Pseudocode Conventions ● Recursive Algorithms 3. Performance Analysis ● Space Complexity ● Time Complexity ● Asymptotic Notations ● Practical Complexities ● Performance Measurements 4. Randomized Algorithms ● Basics of Probability Theory ● Informal Description of Randomized Algorithms ● Identify Repeated Elements ● Primality Testing ● Advantages and Disadvantages
● Review the basic data structures.	Unit 2: Basic Data Structures (2 hours) 1. Stacks and Queues 2. Trees- Binary Trees 3. Dictionaries - Binary Search Tree 4. Priority Queues- Heap and Heapsort 5. Sets and Disjoint Set Union 6. Graphs and Graphs Representation
● Solve problems using divide and conquer strategy	Unit 3: Divide and Conquer Method (5 hours) 1. Introduction 2. Binary Search 3. Finding the Minimum and Maximum 4. Merge sort and Quicksort 5. Selection 6. Strassen's Matrix Multiplication

Solve problems using greedy problem	Unit 4: The Greedy Method (7 hours) - Introduction - Knapsack Problem- Fractional Knapsack Problem - Tree Vertex Splitting - Job Sequencing with Deadlines - Minimum Cost Spanning Trees - Prim's Algorithm - Kruskal's Algorithm - Optimal Storages on Tapes - Optimal Merge Patterns
Solve problems using dynamic programming	Unit 5: Dynamic Programming (9 hours) - Introduction - Multistage Graphs - All Pair Shortest Paths - Single Source Shortest Path: General Weights - Optimal Binary Search Trees - String Editing 0/1 Knapsack - Reliability Design - Traveling Salesman Problem - Flow Shop Scheduling
Use traversal and search techniques in Tree and Graphs	Unit 6: Basic Traversal and Search Techniques (7 hours) - Techniques for Binary Trees - Techniques for Graphs - Breadth First Search and Traversal - Depth First Search and Traversal - Connected Components and Spanning Trees - Biconnected Component and DFS
Solve problems using backtracking strategy.	Unit 7: Backtracking (5 hours) - Introduction - The 8-Queens Problem - Sum of Subsets - Graph Coloring - Hamiltonian Cycle - Knapsack Problem using Backtracking

5. Practical Works

Laboratory works of 45 hours per group of maximum 24 students should cover all the lab works:.

SN	Lab Works
1	Implementation of priority queues, stacks, binary search trees using C++.
2	Implementation of divide and conquer algorithms (quicksort, merge sort) using C++.
3	Implementation of greedy algorithms (Fractional Knapsack problem and Job Sequencing with deadline) using C++.
4	Implementation of dynamic programming (0/1 Knapsack problem and Single Source Shortest Path Problem) using C++.
5	Implementation of graph traversal techniques using C++.
6	Implementation of Sum of Subset Problem using backtracking technique in C++.

6. List of Tutorials:

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (2 hrs)

1. Divide and Conquer Strategy.
2. Greedy Algorithms
3. Dynamic Programming

B. Problem solving-based Tutorials: (10 hrs)

1. Solving recurrence relation related problems using Substitution method , Recursion Tree Method and Master Method.
2. Solving problems of quick sort, merge sort and selection using divide and conquer algorithm.
3. Solving problems of binary search and finding minimum and maximum using divide and conquer algorithms.
4. Solving Fractional Knapsack Problem, Tree Vertex Splitting, Job Sequencing with Deadlines problems using greedy methods.
5. Solving Minimum Cost Spanning Trees, Optimal Storages on Tapes Optimal Merge Patterns using greedy methods.
6. Solving Multistage Graphs ,All Pair Shortest Paths, Single Source Shortest Path, Optimal Binary Search Trees problems using dynamic programming approach.
7. Solving String Editing, 0/1 Knapsack Reliability Design, Travelling Salesman Problem using dynamic programming.
8. Solving Connected Components and Spanning Trees Biconnected Component and DFS problems.
9. Solving Problems using Basic Traversal and Search Techniques.

C. Review and Question/Answer-based Tutorials: (3 hrs)

1. Case study on problem solving using backtracking algorithms followed by Oral Presentation in class.
2. Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation system and Students' Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports and project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books:

1. Horowitz E., Sahni S. & Rajasekaran, S. (1999). *Computer Algorithms C++*. Galgotia Publications.

References:

1. Baase S., and Gelder, A. V. (2012). *Computer Algorithms: Introduction to Design and Analysis*. Addison-Wesley.

Pokhara University
Faculty of Science and
Technology

Course No.: CMP 226 (3 Credits)

Full Marks: 100

Course title: Applied Operating Systems (3-1-2)

Pass Marks: 45

Nature of the course: Theory and Practical

Total Lectures: 45

hrs Level: Bachelor

Program: BE (Software/IT)

1. Course Description

This course is designed to visualize and apply the concepts of operating systems to various system environments. It includes the system software, internal structure, choice and security features of the operating system for running the application software and performing various tasks with optimized throughput. This course also introduces the emerging new trending operating system for distributed environments like cloud systems. This course aims to provide the students with the basic concepts of design and development of an operating system applying necessary security and safety measures.

2. General Objective

- To acquaint the students with the knowledge of the structure of operating systems and their functionality.
- To develop the skills in students to select and design optimal resource allocation schedules.
- To acquaint the students with the knowledge of process and thread, I/Os, Memory, CPU, disk management.
- To acquaint the students with basic concepts of security and safety measures in operating systems.

3. Methods of Instruction

- Lectures, Tutorials, Case Studies, Discussion, Readings and Practical Works.

4. Contents in Detail

Specific Objectives	Contents
- Familiarize with basic concepts of Operating system, and its architecture. - Understanding the successive evolution and structure of operating system	Unit 1. : Introduction [5 hours] 1.1 Applied Operating System 1.2 Goals of an Operating System 1.3 Computer System Organization and Architecture 1.4 Operating-System Structure 1.5 Generations of Operating Systems (0th, First, Second, Third, Fourth) 1.6 Types of Operating Systems 1.6.1 Batch Processing Operating System 1.6.2 Multiprogramming Operating System 1.6.3 Multiprocessing System 1.6.4 Time Sharing 1.6.5 Real Time Operating System (RTOS) 1.6.6 Networking Operating System 1.6.7 Distributed Operating System 1.6.8 Operating Systems for Embedded Devices 1.7 Functions of OS 1.8 Kernel Data Structures 1.9 User and Operating-System Interface 1.10 System Call, Types of System Calls and Handling System Call, 1.11 Virtual Machine
- Familiarize with the concept of Process and threads - Understanding the mutual exclusion for resource utilization - Mastering Process Scheduling Algorithms. - Dealing with Deadlocks and Resource Management:	UNIT- II: Process, Threads and Scheduling [13 hours] 2.1 Process 2.1.1 Process Concept 2.1.2 Process State 2.1.3 Operations on Processes 2.1.4 Process Control Block 2.1.5 Implementation of Processes 2.2 Threads 2.2.1 Thread Concept and Usage 2.2.2 Threads Models (Many-to-one model, One-to-One Model, Many-to many model) 2.2.3 User Space and Kernel Space Threads 2.2.4 Hybrid Implementations 2.2.5 Threads Vs Processes 2.3 Inter-process Communication and Synchronization 2.3.1 Cooperating Processes 2.3.1 Race Conditions 2.3.2 Critical-Section Problem 2.3.3 Mutual Exclusion with Busy Waiting (Disabling Interrupt, Lock variable, Strict alternation, Peterson's Algorithm, Test and Set Lock) 2.3.4 Sleep and Wakeup 2.3.5 Semaphores

	2.3.6 Mutexes 2.3.7 Monitors 2.3.8 Message Passing 2.3.9 Classical IPC Problems 2.4 Process Scheduling 2.4.1 Basic Concept and Types(Preemptive, Non-Preemptive). 2.4.2 Scheduling Criteria 2.4.3 Scheduling Algorithm (First come first served Scheduling(FCFS), Round-Robin,, Shortest Job First(SJF), Shortest process next, Shortest Remaining Time First(SRTF), Priority Scheduling, Priority Fair Share, Real Time Scheduling, Guaranteed, Lottery Scheduling) 2.5 Thread Scheduling 2.6 Deadlocks 2.5.1 System Resources: Preemptable and Non-preemptable 2.5.2 Methods of Handling Deadlock(Deadlock Prevention, Deadlock Avoidance: Banker's Algorithm, Deadlock Detection: Resource allocation graph, Recovery from Deadlock) 2.5.3 Security and Protection- Security Threats, System Protection, Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of Access Matrix, access Control list
• Conceptualize the role and working procedure of memory • Familiarizing with virtual memory management • Understanding the page replacement algorithms • Understand the mechanism file and filing	UNIT- III: Storage Management [12 hours] 3.1 Memory Management 3.1.1 Logical & physical Address Space 3.1.2 Swapping 3.1.3 Contiguous Memory Allocation 3.1.4 Paging, Structure of Page Table 3.1.5 Segmentation, Segmentation with Paging 3.2 Virtual Memory 3.2.1 Background 3.2.2 Demand Paging, 3.2.3 Performance of Demanding Paging, 3.2.4 Page Replacement, Page Replacement Algorithms, Allocation of Frames, 3.2.5 Thrashing. 3.3. File System Interface and Implementation 3.3.1 File System Interface (The Concept of a File, Access methods, Directory Structure, File System Mounting, File Sharing, Protection) 3.3.2 File System Implementation (File System Structure, File System Implementation, and Allocation methods, Free-space Management, Directory Implementation, Efficiency and Performance)

- Understand the role of input/output devices - Understand the different approaches for optimal output	Unit IV : Input/output Management [7 hours] 4.1 I/O System 4.1.1 Principles of I/O Hardware (I/O Device, Device Controller, Memory Mapped I/O, Direct Memory Access) 4.1.2 Principles of I/O Software (Goals of I/O Software, Polled I/O vs Interrupt Driven I/O, Character User Interface and Graphical User Interface, Device Driver, Device Independent I/O Software, User-space I/O Software) 4.2 Mass Storage Structure - Overview of Mass Storage Structure, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, Swap space Management Redundant Array of Inexpensive Disks
Visualize of operating system structure and functionality	Unit V: Case Study [5 hours] 5.1 Linux - Design principles, Inter-process communication, Kernel modules, Network structure, Security 5.2 Windows - Design principles, Programmer interface, System components, Security level
Familiarize with current trends of operating systems	Unit VI : Emerging Trends in Operating System [3 hours] 6.1 Concept, Character and Role of Distributed Operating System and Cloud Operating System 6.2 Security issues and method of deployment 6.3 Memory wall and bottleneck for operating system

5. Practical Works

Laboratory work of 30 hours per group of maximum 24 students should cover the operating system structure and functions of any two popular operating systems. It also insists the students to design a model of operating system with the reference of open source guidelines. Students should complete the following tasks in laboratory:

1. Understanding and running the internal commands and external commands in MS DOS.
2. Installation and user, application management in Windows (current version)
3. Simulation of Process Scheduling Algorithms
4. Simulation of Page Replacement Algorithms
4. Simulation of Disk Arm Scheduling Algorithms
5. System Administration (user, disk, role, etc.) in any open source operating system.

6. List of Tutorials

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (2 hrs)

1. Operating System Services (Class discussion)
2. Deadlock and Starvation
3. New Trends in Operating Systems (Oral Presentation).

B. Problem solving-based Tutorials: (10 hrs)

1. Evaluating the performance of different Process Scheduling Algorithm, Disk Scheduling Algorithm and Page Replacement Algorithm through hypothetical scenarios.
2. WAP that creates multiple processes and manages their execution using different scheduling algorithms such as Round Robin, Priority Scheduling, and Shortest Job First.
3. Implementing page replacement algorithms such as FIFO, LRU, and Clock, and evaluating their performance in a simulated virtual memory environment.
4. WAP that simulates FCFS, SCAN and CSCAN disk scheduling algorithms.
5. Develop basic shell scripts with defined functions, perform simple tests, use loops for iteration, create patterns, utilize expansions, and handle substitutions within shell programming.
6. Develop a program in UNIX utilizing system calls such as fork, exec, getpid, exit, wait, close, stat, opendir, and readdir to perform various operations within the operating system.

C. Review and Question/Answer-based Tutorials: (3 hrs)

1. Case study of “Linux and Windows Operating System” followed by Oral Presentation in class.
2. Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation System and Students’ Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports, project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

9. Prescribed Books and References

Text Books

- 1) A. Silberschatz, P.B. Galvin, G. Gagne. (2009). *Applied Operating System Concepts*. John Wiley & Sons, Inc.

References

- 1) Tanenbaum A.S. (2011). *“Modern Operating System”*, Pearson.
- 2) Dhamdhere D. M. (1999). *System Programming and Operating System*. Tata McGraw-Hill.
- 3) Tanenbaum A. S. (2017). *Distributed Operating System*. Pearson.

Pokhara University
Faculty of Science and Technology

Course No.: CMP 262

Full marks: 100

Course title: **Computer Architecture (3-1-1)**

Pass marks: 45

Nature of the course: Theory & Practical

Total Lectures: 45 hrs

Level: Bachelor

Program: BE (Computer)

1. Course Description

This course is designed to provide the knowledge of the evolution of computer architecture and the factors influencing the design of hardware and software elements of computer systems. It aims to provide an understanding of the design of processing unit and control unit architectures. This course introduces the concepts of instruction set design, processor organization, pipelining, cache and virtual memory organizations, I/O and interrupts, parallel processing and multicore computers.

2. General Objectives

- To acquaint the students with the knowledge of computer architecture and associated processing, control unit and ALU unit of very simple central processing unit.
- To provide the knowledge of the functions of each element of memory hierarchy.
- To develop the skills in students to choose the appropriate Memory and Input Output organization used in real world computing systems.
- To acquaint the students with the knowledge of technology behind modern advanced computer architectures for parallel processing and multicore architecture.

3. Methods of instructions

Lectures, Tutorials, Case Studies, Discussion, Readings and Practical Works.

4. Content in details

Specific objectives	Contents
● Understand the concepts of computer architectures, functional units and components of computer systems and various addressing modes.	Unit 1 Introduction to Architecture [4 Hrs] 1.1. Brief overview of Computer organization and Architecture 1.2. Hierarchy structure of computer system 1.3. Computer evolution and generations 1.4. Computer Components and Functions 1.5. Future Trends in Computer 1.6. Review of Instruction sets, Addressing Modes and Instruction format

Understand the VHDL Programming for simple operations.	Unit -2: Register Transfer Language and Micro operations [4Hrs] 2.1 Register Transfer and RTL 2.2. Micro Operation 2.3 Data Transfer Micro Operations 2.4 Arithmetic and Logical Operations 2.5 Shift Micro operations 2.6 Introduction to HDL and VHDL 2.7 VHDL programming for Adder, Mux, ALU
Understand the functional units of CPU and their organization.	Unit -3: Processor Organization [5 Hrs] 3.1 CPU Organization/Structure 3.2 Register Organization and Data paths 3.3 Instruction Cycle(T states) 3.4 Arithmetic and Logical Unit 3.5 Design Principles for Modern Systems
Understand the design of Hardwired and microprogrammed control units.	UNIT 4 Control Unit [5 Hrs] 4.1 Control of the processor 4.2 Hardwired Control Unit(Control unit inputs/logic) 4.3 Microinstruction Format 4.4 Micro Programmed Control Unit 4.5 Architecture of Microprogrammed Control Unit 4.6 Microinstruction Sequencing and Execution 4.7 Application of Hardwired and Micro programmed Control Units 4.8 RISC and CISC Architecture
Understand the representation of binary numbers in signed and unsigned notation along with the algorithms used for the basic arithmetic operations.	UNIT 5 Computer Arithmetic [7 Hrs.] 5.1 Integer Representation 5.2 Integer Arithmetic 5.3 Unsigned Binary Addition and Subtraction 5.4 Unsigned Binary Multiplication Algorithm 5.5 Booth Multiplication Algorithm 5.6 Unsigned Binary Division Algorithm 5.7 Floating Point Representation
Understand the concepts of pipelining for better performance.	Unit 6: Pipelining [4 hrs] 6.1 Pipelining 6.2 Arithmetic Pipeline 6.3 Instruction Pipeline 6.4 Conflicts in Instruction Pipelining and their solutions 6.5 RISC pipeline 6.6 Register Windowing and Register Renaming
Review memory Hierarchy of computer systems and understand the principles of cache memory to increase the performance of CPU	UNIT 7 Memory Organization [4 Hrs.] 7.1 Memory Hierarchy 7.2 Main Memory and Auxiliary Memory 7.3 Associative Memory and Cache Memory 7.4 Cache mapping techniques- Direct, Associative

	and Set Associative Mapping 7.5 Cache Write Policy. 7.6 Cache Replacement algorithm (FIFO, LRU, LFU)
Familiarize with IO interfaces and introduce various methods for improving I/O performances.	Unit 8: Input-Output Processing [4 Hrs] 8.1 Peripheral Devices 8.2 I/O Modules 8.3 I/O Interface and Techniques 8.4 Modes of Transfer: Programmed, Interrupt-Driven and DMA 8.5 I/O Processor and IO channel 8.6 GPU and TPU 8.7 External Interfaces: FireWire and Infiniband 244
Understand the concept of parallel processing and multi thread architecture in modern processors.	Unit 9: Parallel Processing [4 Hrs] 9.1 Parallel Processing 9.2 Parallelism In Uniprocessor system 9.3 Multiprocessor System and their characteristics 9.4 Flynn Classification 9.5 Interconnection structures in Multiprocessors 9.6 Vector processing and Array processing 9.7 Introduction to Multithreaded Architecture
Prevalent new development in computer architecture: the use of multiple processors on a single chip	Unit 10: Multi-core computer (4Hrs) 10.1 Hardware performance issues 10.2 Software Performance Issues 10.3 Multicore Organization 10.4 Dual Core, Quad Core and Octa Core 10.5 Power Efficient Processor

5. Laboratory Works

Laboratory works of 15 hours per group of maximum 24 students should cover the following lab works:.

1. Write a program to implement the Booth Algorithm (2hr)
2. Write a program to implement the Non Restoring Division Algorithm (2 hr)
3. Write a VHDL Code for Realizing Logic Gates (1hr)
4. Write a VHDL code for Combination circuits (Decoder, Encoder, MUX, Demux ,Comparator, Code converter) (6 Hrs)
5. Write a VHDL Program for Realizing Sequential Circuits Like Fliflop and counters (4 Hrs)

6. List of Tutorials:

The various tutorial activities that outfit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of

the instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 15 hrs should be conducted to cover all the content of this course:

A. Discussion based Tutorials (3 hrs)

1. Overview of Computer Organization and Architecture.
2. Memory hierarchy for modern processors.
3. Comparative analysis of different aspects of computing systems as defined in Flynn's Classification.

B. Problem Solving based Tutorials (6 hrs)

1. Design a CPU for any given registers set, instruction set and state diagram.
2. Develop a control unit for any given state diagram.
3. Design a microsequencer control unit for any given specifications following design procedure.
4. Perform arithmetic addition and subtraction in signed and unsigned notation for any given numbers.
5. Perform Multiplication operation for any given numbers using shift-add multiplication algorithm and Booth's algorithm.
6. Perform Division operation for any given numbers using restoring and Non restoring Division algorithm.

C. Review and Question/Answer-based Tutorials: (6 hrs)

1. Case study on any of multi threaded and Multi core processors. It should include the architecture of processor, control unit, memory as well as input output organization in detail. An oral presentation with the submission of a report should be a part of work and must be included as a component for evaluation.
2. Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation system and Students' Responsibilities

Evaluation:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30		
Attendance and class Participation	10%			
Assignments	20%			
Project work/Presentations	20%			
Term Exam	50%			
Practical		20		

Attendance and Lab Participation	20%		Semester End examination	50
Lab report	30%			
Practical Exam	30%			
Viva	20%			
Total Internal Marks		50		
Full marks=50+50				

Students Responsibility:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the semester End Examination. Failing to get such a score will be given NOT QUALIFIED(NQ) to appear for the Semester End Examination. Students are advised to attend all the classes, formal exam, test and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Text Books and References

Text Books:

1. Stalling W. (2011). *Computer Organization and Architecture*, Pearson Education.

References:

2. Carpinelli, John D. (2001). *Computer System Organization and Architecture*. Pearson Education Asia.
3. Hall, Douglas V. (2005). *Microprocessor and Interfacing programming and Hardware*. McGraw Hill, New Delhi.
4. Tanenbaum, A.S. (2003). *Structured Computer Organization*. Pearson Education.
5. Uffenbeck, J. (1991). *Microcomputers and microprocessors: the 8080, 8085, and Z-80 programming, interfacing, and troubleshooting*. Prentice-Hall, Inc..
6. Moris M. M. (1992). *Computer System Architecture*. Pearson

Pokhara University
Faculty of Science and Technology

Course Code: CMP 274

Full marks: 100

Course title: **Computer Graphics and Multimedia (3-1-2)**

Pass marks: 45

Nature of the course: Theory/Tutorial/Practical

Total periods: 45

Level: Bachelor

Program: BE SE

1. Course Description

This course is designed to provide students with an in-depth knowledge of representations of graphical objects, scenes and techniques of multimedia system development. The course provides the concepts of graphical hardware, two dimensional and three dimensional representations of graphical objects. In addition, this course provides the knowledge of the tools and techniques of multimedia systems.

2. General Objectives

- To familiarize the students with the technologies used for representing and rendering of graphical objects and scenes in a computer system
- To equip the students with necessary skills and knowledge of two dimensional and three dimensional graphics.
- To provide the students with the knowledge of algorithms and approaches used for visual realism in the computer generated graphical scenes and objects.
- To familiarize students with the technologies associated with multimedia system development.

3. Methods of Instruction

Lectures, Tutorials, Case Studies, Discussion, Readings and Practical/Project Works.

4. Contents in detail with specific objectives

Specific Objectives	Contents
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Familiarize with basic graphics hardware and software.	Unit 1: Computer Graphics (4 hrs) 1.1 Introduction 1.2 Raster and Random Scan Systems 1.3 Video Controller and Frame Buffer 1.4 Graphics Software 1.5 Graphics Hardware and Graphics Cards (GPUs) 1.6 Recent trends in Computer Graphics
Implement the scan conversion algorithms using a programming language.	Unit 2: Scan Conversion Algorithms (5 hrs) 2.1 Scan Conversion 2.2 Line Drawing Algorithms 2.2.1 Digital Differential Analyzer 2.3.2 Bresenham's Line Algorithm 2.3 Circle Generation Algorithm
Use the various transformation and viewing operations to reposition and resize objects in two dimensional scene	Unit 3: Two Dimensional Geometric and Viewing Transformations (6 hrs) 3.1 Two Dimensional Transformations (Translation, Rotation, Scaling) 3.2 Matrix Representations of Transformations 3.3 Composite Transformations 3.4 Two Dimensional Viewing Pipeline 3.5 Window to Viewport Transformation 3.6 Line Clipping Algorithm: Cohen-Sutherland
Apply the three dimensional rendering and transformations for representing objects more accurately	Unit 4: Three Dimensional Graphics (5 hrs) 4.1 Three Dimensional Transformations - Translation, Rotation and Scaling 4.3 Three Dimensional Representations - Polygon Surfaces and Solid Modeling 4.3 Projections- Parallel and Perspective
Develop the programs using functions in OpenGL library for drawing graphical output primitives and performing basic transformations	Unit 5: Visual Realism and OpenGL (8 hrs) 5.1 Hidden Surface Removal Approaches - Back-Face Detection, Depth Buffer Method and Scan Line Method 5.2 Illumination Theory and Models - Ambient Light, Diffuse Reflection and Specular Reflection 5.3 Shading Methods - Constant Shading, Gouraud Shading and Phong Shading 5.4 OpenGL and its Architecture, Drawing Basic Output Primitives and Polygons, Call Back Functions and Input Handling, Basic Transformations, Projections and Lighting

● Understand the basic concepts of multimedia systems and media representation methods.	Unit 6: Multimedia System and Media Representation (7 hrs) 6.1 Multimedia System 6.1.1 Definition, Scope and Properties 6.1.2 Continuous and discrete media 6.1.3 Evolution and trends in multimedia technology 6.1.4 Analysis of common image formats (BMP, JPEG and PNG) 6.1.5 Applications of multimedia systems 6.2 Sound and Audio 6.2.1 Digital audio formats (WAV and mp3) 6.2.2 Audio sampling and quantization 6.2.3 Frequency, amplitude and waveform 6.2.4 MIDI hardware, software and messages 6.3 Video: 6.3.1 Digital video formats (AVI and MOV) 6.3.2 Video codecs, frame rate, resolution and color depth 6.4 Animation: 6.4.1 Hand drawn, stop motion and computer animation 6.4.2 Morphing 6.4.3 Basic steps in animation
● Understand and apply the fundamentals of data compression and coding techniques. ○	Unit 7: Multimedia Compression Techniques (6 hrs) 7.1 Image Compression - JPEG compression standard 7.2 Psychoacoustic principles, perceptual audio coding, Audio compression technique - Lossy audio compression algorithms (MP3 and AAC) 7.3 Video compression standards - Analysis of video compression standards (MPEG, MPEG-2, MPEG-4, and H.264/AVC)
● Understand the multimedia authoring tools and user interface design principles for multimedia applications.	Unit 8: Multimedia Application Development (4 hrs) 8.1 Multimedia authoring tools and frameworks 8.2 User interface design for multimedia applications 8.3 Introduction to Multimedia Workstation 8.4 Real time requirements in multimedia systems

5. List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted:

A. Discussion-based Tutorials: (4hrs)

1. Raster and random display technology.
2. Parallel and Perspective Projection.
3. Significance of MIDI hardware, software and messages.
4. Presentation and representation of image in B/W, grayscale and color

B. Problem solving-based Tutorials: (5 hrs)

1. Digitize the line with endpoints A (18,30) and B (10,20) using DDA and BLA
2. Rotate the triangle with vertices A (0, 0), B (1, 1) and C (5, 2) by angle -90 degrees in clockwise direction about X=2 line.
3. You are provided with the clipping rectangle with coordinates A (10,10), B (10,20), C (20,10) and D(20,20). Clip a line with endpoints P (5,15) and Q (25,15) with the clipping rectangle using Cohen-Sutherland algorithm.
4. The pyramid defined by the coordinates A(0,0,0), B(1,0,0), C(0,1,0) and D(0,1,0) is scaled to double of its size about point (0,1,0) then translated to new position by translation vectors (4,5,6) and rotated by angle 45 degree in clockwise direction about origin. Use a homogeneous transformation matrix to find the new vertices of the pyramid.
5. Calculate the total time required to transmit a minute long video of spatial resolution 640x480, 32 bits per pixel and 12 frames per second through a communication link at a constant rate of 56 Kbps

C. Review and Question/Answer-based Tutorials: (6 hrs)

1. Case study on “**Game development using OpenGL**” followed by Oral Presentation in class.
2. Case study on “**Designing User Interfaces for Multimedia Applications**” followed by Oral Presentation in class. You may choose any one multimedia application such as video conferencing, multimedia streaming, online multiplayer gaming etc.
3. Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

6. Practical Work

Laboratory works of 30 hours per group of maximum 24 students should cover the following lab works:

S.N.	Practical works
1	To draw a straight line using Digital Differential Analyzer and Bresenham's Line Drawing Algorithm
2	To digitize a circle using Midpoint Circle Drawing Algorithm
3	To draw Two Dimensional objects on the screen and perform two dimensional transformations to them
4	To draw the projected view of a Three Dimensional object using perspective and parallel projection and perform three dimensional transformations to it
5	To draw graphical output primitives using OpenGL, perform various transformations and implement algorithms covered in the course
6	To implement and test the Image Compression algorithm (JPEG Compression)
7	To perform audio compression analysis and conversion (MP3 vs. AAC)
8	To familiarize with Video Compression Standards: MPEG-2, MPEG-4, H.264/AVC and conversion from one another
9	To create a simple 2D and 3D animation using HTML, CSS and JS

Students are required to demonstrate the assimilated knowledge in Computer Graphics and Multimedia through mini-project work (Games, Graphical Simulations, Algorithm demonstration) and need to submit at the end of the semester. Students are allowed to explore new programming languages or platforms such as Unity, Unreal Engine or any related tools for accomplishing the project work.

7. Evaluation system and students' responsibilities

Internal Evaluation

In addition to the formal end-semester exam(s), the internal (formative) evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation and presentation etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30		
Attendance & Class Participation	10%			

Assignments	20%		Semester-End examination	50
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	20%			
Lab Report/Project Report	30%			
Mini-Project	50%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student requirements:

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the semester-end examination. Failing to get such a score will be equated with NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Semester examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. Failure of a student to attend a formal exam, quiz, test, etc. won't qualify him/her for re-exam. ***Students are required to complete all the requirements defined for the completion of the course***

8. Prescribed Books and References

Text Book

1. Hearn, D. & Baker M. P. (1997). *Computer graphics, C version*. Pearson Education India.
2. Buford, J. F. K. (2002), *Multimedia Systems*, Pearson, India

Reference Books

1. Hearn, D. & Baker M. P. (2004). *Computer graphics with OpenGL*. Pearson Education India.
2. Foley J. D, Dam A. v, Feiner & S. K. & Hughes J. F. *Computer Graphics: Principles and Practice in C*. Addison-Wesley

3. Woo M., Neider J., Davis T. & Shreiner D. *OpenGL Programming Guide: The Official Guide to Learning OpenGL*. LPE Pearson Edition Asia
4. Ralf Steinmetz and Klara Nahrstedt,
5. Steinmetz R. and Nahrstedt K.. *Multimedia Systems*.
6. Ralf Steinmetz, Klara Nahrstedt, and Thomas Sikora, .
7. Steinmetz R. and Nahrstedt K. & Sikora T. (1995). *Multimedia: Computing, Communications & Applications*.
8. Ze-Nian Li, Mark S. Drew, and Jiangchuan Liu, . Prentice Hall.
9. Li Z., Drew M. S., & Liu J. *Fundamentals of Multimedia*..Springer.

Pokhara University
Faculty of Science and Technology

Course No.: CMP 278

Full marks: 100

Course title: Object Oriented Design and Modeling using UML (3-1-2)

Pass marks: 45

Nature of the course: Theory & Practical

Total Lectures: 45 hrs.

Level: Bachelor

Program: BE (Software)

1. Course Description

This course is designed to provide the fundamental concepts of object oriented analysis and design of software systems using the tools of Unified Modeling Language. This provides the students with knowledge of how design of software processes starts and how the UML is used in modeling every aspect of the whole development process.

2. General Objectives

- To familiarize the students with the fundamental concepts of Object Oriented Analysis and Design.
- To provide students with the knowledge of Object-Oriented Modeling language- the Unified Modeling Language (UML).
- To develop the skills in the students to design the software system in object-oriented framework using UML.

3. Methods of Instruction

Lectures, Tutorials, Case Studies, Discussion, Readings and Practical Works.

4. Contents in Detail.

Specific Objectives	Contents
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● Understand the basic Object-Oriented concepts.	Unit 1: Object Oriented Fundamentals (10 hours) 1.1. Introduction 1.2. Object Oriented Analysis and Design 1.3. Overview of UML: Fundamentals and Notations 1.4. Review of Software Process Models ● Waterfall, Iterative, Evolutionary Models ● Agile Model 1.5. Unified Process 1.6. Inception and Use Cases 1.7. Use Case Diagrams and Activity Diagrams 1.8. Object Oriented Development Cycle
● Analyze systems and develop models using object-oriented approaches.	Unit 2: Object Oriented Analysis (12 hours) 2.1. Building Conceptual Model 2.2. Adding Associations and Attributes 2.3. Representation of System Behaviors 2.4. Package Diagrams 2.5. Developing Class Diagrams from Domain Model
● Design systems using an object-oriented approach.	Unit 3: Object Oriented Design (14 hours) 3.1. Analysis to Design 3.2. Describing and Elaborating Use Cases 3.3. Interaction Diagrams ● Sequence Diagrams ● Communication Diagrams 3.4. Objects and Patterns ● Introduction to GRASP Pattern 3.5. Determining Visibility 3.6. Updating Class Diagrams 3.7. Concept of Object Diagrams and State Diagrams
● Develop and implement code from design documents.	Unit 4: Implementation (9 hours) 4.1. Programming and Iterative, Evolutionary Development 4.2. Mapping Design into Codes 4.3. Creating Class Definition from Design Class Diagrams 4.4. Creating Methods from Interaction Diagrams 4.5. Collection Classes in Code 4.6. Exceptions and Error Handling 4.7. Order of Implementation 4.8. Introduction to Test Driven Development 4.9. Concept of Deployment Diagram and Component Diagram

5. Practical Works

Laboratory works of 30 hours per group of maximum 24 students should cover the following lab works:

SN	Lab Works
1	Develop Use Cases and Use Case diagrams
2	Develop Domain Model and Class diagrams
3	Develop Activity and State diagrams
4	Develop Interaction diagrams (Sequence and Communication diagrams)
5	Develop Code from given class and interaction diagrams

6. List of Tutorials:

Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (2 hrs)

1. Object Oriented Analysis and Object Oriented Design (Class discussion)
2. Unified Modeling Language and its Applications.

B. Problem solving-based Tutorials: (10 hrs)

1. Develop Use Cases and Use Case diagrams
2. Develop Domain Model and Class diagrams
3. Develop Activity and State diagrams
4. Develop Interaction diagrams (Sequence and Communication diagrams)
5. Develop Code from given class and interaction diagrams.

C. Review and Question/Answer-based Tutorials: (3 hrs)

1. Consider a real-life problem and model them using UML tools. Prepare and submit complete documentation followed by oral-presentation.
2. Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation system and Students' Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports and project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books:

1. Larman, C. (2012). *Applying UML and patterns: an introduction to object oriented analysis and design and Unified Process*. Pearson Education India.

References:

1. Stevens, P. Pooley, R. (1999), *Using UML: Software Engineering with Objects and Components*, Addison-Wesley.
2. Fowler, M. Scott, K. (1997), *UML Distilled: Applying the Standard Object Modeling Language*, Addison-Wesley.

3. Miles, R., Hamilton, K. (2006), *Learning UML 2.0: A Pragmatic Introduction to UML*, O'Reily Media Inc.

Pokhara University
Faculty of Science and Technology

Course No.: CMP 270

Full marks: 100

Course title: **Research Fundamentals (2-0-2)**

Pass marks: 45

Nature of the course: Theory & Practical

Total Lectures: 45 hrs

Level: Bachelor

Program: BE (Computer, IT and Software)

1. Course Description

This course is designed to develop the skills of students to do a project/research work using the fundamental concepts of research. This course introduces what the research or project work is, explores in brief how it is done, explains what research ethics must be followed during the research/project work and finally guides the students how the research/project documentation is done in the form of the proposals report and final research/project report.

2. General Objectives

- To acquaint the students with basic knowledge of research/project work.
- To develop the skills in students to conduct research/project work.
- To develop the skills in students to work in a team.
- To develop the skills in students to write an impressive proposal report and final research/project report and present their work orally.
- To acquaint the students with the knowledge of research ethics.

3. Methods of Instruction

Lecture, Discussion and Project work.

4. Contents in Detail.

Specific Objectives	Contents
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Understand the basic concepts of research, purpose and outcomes of a research/project work.	Unit 1: Introduction (4 hrs) 1.1 What is research? 1.2 Research Aim and Objectives 1.3 Features of Research 1.4 Types of Research 1.5 The 6Ps of Research 1.6 Purpose of Research- reasons for doing research 1.7 Product of Research- outcomes of research 1.8 Research and Project
Understand and implement the research process model to conduct a research/project work.	Unit 2: Research Process Model (10hrs) 2.1 Personal Experiences and Motivation 2.2 Literature Review - Purpose and objectives of a literature review - Literature resources - Conducting a literature review - Citation and its types - Bibliographic Detail and Referencing Systems - Plagiarism 2.3 Research Question 2.5 Conceptual framework 2.5 Strategies - Survey - Design and Creation - Experiment - Case Study - Action Research and - Ethnography 2.6 Data Generation Methods - Interview - Observations - Questionnaire - Documents - Types of triangulation in a research project 2.7 Data Analysis - Quantitative and - Qualitative data analysis
Familiarize with the laws and ethics in research conduction.	Unit 3: Participants and Research Ethics (4hrs) 3.1 Participants 3.2 The law and Research 3.3 Rights of People Directly Involved 3.4 Responsibilities of an Ethical Researcher

• Familiarize with the research proposal and its components. • Develop a research/project proposal.	Unit 4: Proposal Writing (4hrs) 4.1 What is a research proposal? 4.2 Need of a Research Proposal 4.3 Components of a Research Proposal 4.4 A case study on any research paper/project
• Familiarize with the research/project report and its components. • Develop a research/project report.	Unit 5: Report Writing (4hr) 5.1 What is a research report? 5.2 Need of a Research Report 4.3 Components of a Research Report 4.4 A case study on any research paper/report

5. Practical Work

Laboratory works of 30 hours per group of maximum 24 students should cover all the concepts of research fundamentals studied in the lectures. Students must find a new problem, write a proposal, solve the problem as their project/research work and submit a final project/research report and present their work orally. The marks for the practical work will be based entirely on their project/research work. The entire project/research work shall be divided into two phases and evaluation shall be done accordingly:

Phase I:

- The students are grouped in teams each containing at most 4 students.
- Each team chooses a problem to solve as their project/research work and they work in a team.
- They must define clearly what the problem is, justify why they choose the problem and how they will solve it and submit this as a proposal report (based on Unit 2 and 4).
- Each team presents their proposal orally.

Phase II:

- After the approval of their proposal, they start working on the project.
- Each team follows the research process studied in Unit 2 to do their project/research work.
- Students keep reporting their progress about the project/research work to their instructor.
- Students complete the project/research work, develop the final project/research report (based on Unit 2 and 5) and again present it orally.

6. Evaluation system

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30		
Attendance & Class Participation	10%			
Assignments	20%			

Presentations/Quizzes	10%		Semester-End examination	50
Internal Assessment	60%			
Practical (Project/research Work)		20		
Proposal Report	30%			
Project Presentation	10%			
Final Project Report	40%			
Completeness of Project	20%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

7. Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

1. Oates, B. J., Griffiths, M., & McLean, R. (2022). *Researching information systems and computing*. Sage.
2. Walia, A. M., & Uppal, M. (2020). *Fundamentals of Research*. Notion Press.

8. Annex

A. Components of Research Proposal

1. Title Page
2. Abstract
3. Table of Contents, List Figures, List of Tables and Abbreviations
4. Introduction
 - a. Rationale/background
 - b. Problem and motivation
 - c. Aim and objectives of research
 - d. Significance of research

- e. Scope of research
 - f. Limitation
- 5. Literature review
- 6. Research problem and Solution
- 7. Methodology
 - a. Research design
 - b. Participants
 - c. Data collection methods
 - d. Data analysis techniques
 - e. Ethical considerations
 - f. Validation Techniques
- 8. Data Analysis and Findings
- 9. Discussions and Conclusion
- 10. Contributions and Future Works
- 11. Reference list/bibliography
- 12. Annexes

B. Components of Research Report:

- 1. Title
- 2. Abstract
- 3. Keywords
- 4. Rationale/background and motivation
- 5. Aim and objectives of research
- 6. Literature review
- 7. Research problem
- 8. Methodology
- 9. Research plan and budget
- 10. Contributions- impact and significance
- 11. Reference list/bibliography
- 12. Annexes

Other Parts of Research Report:

- 1. Funding and Acknowledgement
- 2. Table of Contents, List Figures, List of Tables and Abbreviations
- 3. Title Page and Copyright Page
- 4. Declaration and Recommendation
- 5. Certification Page

***Note:** The components of research proposal and research report may vary based on the research strategy and nature of the research problem.*

Pokhara University
Faculty of Science and Technology

Course No.: CMP 268 (2 Credit)

Full marks: 100

Course title: System Administration and IT Infrastructure Services (**2-1-2**)

Pass marks: 45

Nature of the course: Theory & Practical

Total Lectures: 30 hrs.

Level: Bachelor

Program: BE (IT)

1. Course Description

This course is designed to provide the students with the theoretical knowledge and practical skills for effective system administration and IT infrastructure management. Effective and secure IT systems are essential for the success of enterprises across a range of industries in today's technology-driven environment. This course intends to develop theoretical understanding and practical skills in students to work as system administrators and IT infrastructure professionals. Through this course, students learn about Linux and Windows Server administration, virtualization technologies, and cloud computing.

2. General Objectives

- Acquaint the students with knowledge of the role and responsibilities of system administrators and their significance in IT infrastructure management.
- Provide the students with the concepts of virtualization technologies and their applications.
- Acquaint the students with skill in administering both Linux and Windows server administration for IT infrastructure management.
- Provide the basic concepts of Cloud Computing to understand cloud computing principles, services, and deployment strategies.

3. Methods of Instruction

Lecture, Discussion, Readings, Practical works, Case Studies and Project works.

4. Contents in Detail.

Specific Objectives	Contents
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Understand the basics of System Administration, IT Infrastructure components and necessity of IT Service management for System Admins.	Unit 1: Introduction to System Administration & IT Infrastructure Services (6 Hours) 1.1 Introduction to System Administration 1.2 Role and Responsibilities of a System Administrator 1.3 Importance of System Administration in IT Infrastructure 1.4 History and Evolution of System Administration 1.5 IT Infrastructure Components (Hardware and Software) 1.6 Types of IT Services (Web, Email, Database, etc.) 1.7 Service Level Agreements (SLAs) 1.8 IT Service Management (ITSM) Frameworks
Understand Linux and Windows Operating System in terms of System Administration Perspective	Unit 2: Basics of Linux and Windows System Administration (6 Hours) 2.1 Introduction to Linux Distributions (e.g., Ubuntu, CentOS) 2.2 Basic Linux Commands and File System Structure 2.3 User and Group Management in Linux 2.4 Linux File Permissions and Security 2.5 Introduction to Windows Server Operating Systems 2.6 Active Directory and Domain Controllers 2.7 Windows Server Roles and Features 2.8 Windows Server Security and Group Policies
Understand and implement the networking protocols and security concepts.	Unit 3 : Network Fundamentals and Security in IT Infrastructure Services (7 Hours) 3.1 Basics of Networking Protocols 3.2 Fundamentals of network reference models.(TCP/IP,OSI) 3.3 IP Addressing 3.4 Network Topologies and Models Fundamentals of DNS, Web and File Server 3.5 Principles of Information Security 3.6 Common Security Threats and Vulnerabilities 3.7 Security Best Practices for Servers and Networks
Understand the basic concepts of virtualization and cloud computing.	Unit 4 : Virtualization and Cloud Computing (5 Hours) 4.1 Virtualization vs. Physical Infrastructure 4.2 Types of Virtualization (Server, Desktop, Network, Storage) 4.3 Virtualization Platforms (E.g. VMware, Virtual Box, Hyper-V) 4.4 Introduction to Cloud Computing 4.5 Cloud Computing Services (SPI Model)

● Gain the skill to effectively use scripting languages, configuration tools, and troubleshoot in varied IT environments.	Unit 5 : Automation and Scripting for System Administration and Troubleshooting (6 Hours) 5.1 Scripting Languages (e.g., Bash, PowerShell) 5.2 Automating Routine Tasks with Scripts 5.3 Configuration Management Tools (e.g., Ansible) 5.4 DevOps Principles and Practices 5.5 Troubleshooting Methodology 5.6 Case Studies in Troubleshooting 5.7 Remote Administration
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5. Practical Works

The practical works of 30 hours per group of maximum 24 students should cover the following lab works:

SN	Practical Works
1.	Install and Configure a Linux Server. Setup DNS, Web Server and File Server
2.	Install and Configure a Windows Server. Setup DNS, Active Directory and Web Server
3.	Virtualization Lab. Install ESXI and create different Virtual Machines
4.	Cloud Computing Lab with AWS on EC2, VPC, S3, Auto scaling
5.	Linux and Windows Scripting LAB with Bash and PowerShell

6. List of Tutorials:

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (6 hrs)

1. Role of IT Service Management for System Administration and IT Infrastructure Services.
2. User and Group Management in Windows.
3. User and Group Management in Linux Systems.

B. Problem solving-based Tutorials: (4 hrs)

1. Identify Security Best Practices for Servers and Networks
2. Identify the Benefits of DevOps in System Administration

C. Review and Question/Answer-based Tutorials: (5 hrs)

1. Case study on “Merits and Demerits of Cloud Computing Services” followed by Oral Presentation in class.

- Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation system and Students' Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports and project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books:

1. Nemeth, E., Snyder, G., Hein, T. R., & Whaley, B. (2020). *UNIX and Linux System Administration Handbook*. Addison-Wesley Professional.
2. Panek C. (2019). *Windows Server Administration Fundamentals*. Wiley.

References:

1. Blum, R. (2020). *Linux Command Line and Shell Scripting Bible*. Wiley.
2. Agutter, C. (2013). *ITIL Foundation Essentials: The Exam Facts You Need*. IT Governance Publishing.

Pokhara University
Faculty of Science and Technology

Course No.: CMP280

Full marks: 100

Course title: **Systems Programming (3-1-2)**

Pass marks: 45

Nature of the course: Theory and Practical

Total Lectures: 45 hrs

Level: Bachelor

Program: BE (Software)

1. Course Description

This course provides the basic concepts of various components essential for the both machine-dependent and machine-independent systems, such as assemblers, loaders, linkers, microprocessors, and certain compiler elements. Additionally, this course intends to provide the students with the knowledge to develop an assembler, linker, loader, and compiler.

2. General Objectives

- To acquaint the students with the knowledge of the hypothetical and traditional machine architectures.
- To provide the students with the knowledge of the basic structure and design of an assembler, a linker, a loader, and a compiler.
- To acquaint the students with basic concepts to design and implement system softwares to support the various architectures of machines.
- To develop the skills in students to design and implement macro-processor.
- To provide the students with the basic concepts of compiler design and object oriented design of assembler.

3. Methods of Instruction

Lecture, Discussion, Readings, Practical works.

4. Contents in Detail.

Specific Objectives	Contents
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- To understand SIC and SIC/XE architecture. - To enable students to write and understand SIC programs. - To explain the concept of RISC and CISC	Unit 1: Introduction to Machine Architectures (5 Hrs.) 1.1. System software and machine architecture 1.2. SIC and SIC/XE architectures, SIC Programs. 1.3. RISC and CISC machine architectures.
- To understand the functions of assembler. - To review algorithms and data structures used in assembler design - To demonstrate machine independent and machine dependent features of assembler. - To trace one-pass two-pass assembler design.	Unit 2: Assembler Design (10 Hrs.) 2.1. Basic assembler functions: Simple SIC Assembler, Algorithms and Data structures for Assembler 2.2. Machine dependent assembler features: Instruction formats and addressing modes, program relocation 2.3. Machine independent assembler features: Literals, Symbol defining statements, program blocks, control sections, program linking 2.4. Assemblers Design Options: One pass assembler, multi-pass assembler
- To understand functions and linkers and loaders - To explain machine dependent and machine independent features of loader - To demonstrate working of linker - To trace working of loader	Unit 3: Loader and Linker Design (10 Hrs.) 3.1. Basic loader functions: Design of absolute loader, Simple Bootstrap loader 3.2. Machine dependent features: Relocation, Program Linking 3.3. Machine independent Loader Features: Automatic library Search, loader options 3.4. Design Options: Linking Loader, Linkage Editor, Dynamic Linking

- To understand the concept of macro and macro processor - To explain machine dependent and machine independent features of macro processor - To trace working of macro processor - To demonstrate macro processor design options	Unit 4: Macro Processor Design (10 Hrs.) 4.1. Basic Macro processor functions: Macro Definition and Macro Expansion, Data structures for Macro processor 4.2. Machine Independent features: Concatenation of Macro Parameters, Generation of Unique Labels, Conditional Macro Expansion 4.3. Macro processor Design Options: Recursive Macro Expansion, General Purpose Macro Processors
- Main phases of compiler - Demonstrate working of lexical analyzer - Trace working of parser	Unit 5: Fundamentals of Compiler Design (5 Hrs.) 5.1. Compiler Functions: Grammar, Lexical Analysis, Syntactic Analysis. 5.2. Operator precedence parsing, Recursive Descent, Intermediate Code Generation.
- Understand the basic object oriented design concept of the assembler.	Unit 6: Object-Oriented Design of Assembler (5 Hrs.) 6.1 Booch's Macro Processes 6.2 Booch's Micro Processes 6.3 Object Diagram of Assembler 6.4 Interaction Diagram of Assembler

5. Practical Works

Laboratory works of 30 hours per group of maximum 24 students should cover all contents studied in this course. Students should submit lab sheets (or a final project that uses the knowledge studied in this course) for evaluation.

List of Practical:

SN	Practical Topic
1	Project 1: Symbol table management
2	Project 2: Handling of literals and expressions
3	Project 3: First Pass of assembler design
4	Project 4: Second Pass of assembler design

5	Project 5: Linking Loader design
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6. List of Tutorials:

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (6 hrs)

1. Instruction and addressing modes.
2. SIC/XE program features.
3. SIC/XE assembler design.
4. Object code generation.
5. Macro processor design
6. Design of Linking Loader.

B. Problem solving-based Tutorials: (6 hrs)

1. Writing SIC/XE programs
2. Tracing first pass of assembler
3. Tracing second pass of assembler
4. Tracing program linking
5. Tracing linking loader
6. Tracing first and second pass of macro processor

C. Review and Question/Answer-based Tutorials: (3 hrs)

1. Case study of “Macro Processor” followed by Oral Presentation in class.
2. Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation system and Students’ Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports and project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory	30	50		

Attendance & Class Participation	10%		Semester-End examination	50
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical	20			
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal	50	50		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books:

1. Beck, L. L. (1997). *System software: an introduction to systems programming*. Pearson Education India.

References:

1. Marcellus, D. H. (1984), *Systems programming for Small Computers*, Prentice-Hall.
2. Ellzey, R. S. (1987), *Computer System Software: The Programmer/Machine Interface*, Science Research Associates.
3. Robert M. Graham (1975), *Principles of Systems Programming*, John Wiley & Sons.
4. Thomas G. Windeknecht (1983), *Systems programming*, Little Brown.

Pokhara University
Faculty of Science and Technology

Course No.: CMP264

Full marks: 100

Course title: **Theory of Computation (3-1-0)**

Pass marks: 45

Nature of the course: Theory

Total Lectures: 45 hrs

Level: Bachelor

Program: BE (Computer)

1. Course Description

This course is designed to provide basic knowledge of the theory of automata, formal languages and computational complexity.

2. General Objectives

- To acquaint the students with the basic knowledge of automata and formal languages.
- To develop the skills in students to design various types of automata and analyze them.
- To acquaint the students with the concepts of computability, computational bounds and computational complexity.

3. Methods of Instruction

Lecture, Discussion, Readings, Tutorials.

4. Contents in Detail.

Specific Objectives	Contents
- Understand the concept of alphabet and language	Unit 1: Introduction (4 hrs.) 1.1 Review of set, relation and function 1.2 Proof techniques– proof by contradiction, pigeon hole principle, induction and diagonalization. 1.3 Alphabets and language 1.4 Chomsky's hierarchy?

- Design and implement the deterministic and non-deterministic finite automata. - Develop the equivalence of regular languages and finite automata	Unit 2: Finite Automata and Regular Language (10 hrs) 2.1 Deterministic Finite Automata, Non-Deterministic Finite Automata 2.2 Regular expressions and regular language, equivalence of regular language and finite automata 2.3 Properties of regular language 2.4 Pumping lemma for regular sets 2.5 Closure properties of regular sets 2.6 Decision algorithms for regular sets
- Explain the theory and design of context-free grammar and pushdown automata and their equivalence. - Explore the derivation trees, simplification and formal forms of context-free grammar.	Unit 3: Context-Free Language and Pushdown Automata (13 hrs) 3.1 Context-free grammar 3.2 Derivative trees and simplification of context-free grammar 3.3 Normal forms (CNF, GNF) 3.4 Pushdown automata (formal description and final state PDA design) 3.5 Equivalence of pushdown automata and context-free grammar 3.6 Properties of context-free languages (CFL) 3.7 Pumping lemma for CFL's 3.8 Closure properties of CFL's 3.9 Decision algorithms for CFL's
- Explain the theory and significance of Turing machines - Explain computing mechanism and extensions of Turing machines - Understand the computable languages, functions and unrestricted grammar	Unit 4: Turing Machines (10 hrs) 4.1 Introduction to Turing machine 4.2 Computing with Turing machine 4.3 Extensions of Turing machine 4.4 Unrestricted grammar 4.6 Recursively enumerable languages
- Use the idea of undecidability introducing Church-Turing thesis and the halting problem - Understand the universal Turing machines and undecidable problems of Turing machines	Unit 5: Undecidability (4 hrs) 5.1 The Church-Turing thesis 5.2 Halting Problem 5.3 Universal Turing machines 5.4 Undecidable problems about Turing machines 5.5 Properties of Recursive and Recursively enumerable languages.

- Understand the concept of computational complexity and different classes of problems	Unit 6: Computational Complexity (4 hrs) 6.1 Introduction to Complexity theory, tractable and intractable problems. 6.3 Class P and Class NP problems 6.4 NP-complete problems.
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5. List of Tutorials:

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (2 hrs)

1. Review discussion on set theory concepts like functions, relations, etc.
2. Discussion on the theoretical meaning and significance of computation and computers
3. A study and presentation can be done on the history of computation and Alan Turing as the father of computation (Oral Presentation).

B. Problem solving-based Tutorials: (8 hrs)

1. Solve problems to design deterministic and non-deterministic finite automata (DFA and NDA) to recognize/generate given regular languages.
2. To develop regular expressions to recognize/generate given regular languages.
3. To convert a given NDA to DFA.
4. To design Context-Free Grammar (CFG) to recognize/generate given Context-Free Language (CFL)
5. To simplify a given CFG and convert it into CNF and GNF
6. To design Pushdown Automata (PDA) to recognize/generate given Context-Free Language (CFL)
7. To design Turing machines to recognize/generate given languages

C. Review and Question/Answer-based Tutorials: (5 hrs)

1. Case study of examples of DFA, PDA and Turing Machines for some practical tasks followed by Oral Presentation/demonstration in class.
2. Case study of Unsolvable problems such as the Tiling problem followed by Oral Presentation in class.
3. Case study of Class NP and NP-complete problems (eg, Travelling Salesman Problem) followed by Oral Presentation in class.

7. Evaluation system and Students' Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment, etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		50	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books:

1. Lewis, H. R., & Papadimitriou, C. H., *Elements of theory of computation*, Pearson Education.
2. R. McNaughton, *Elementary Computability, Formal Languages, and Automata*, Prentice Hall of India.
3. Hopcroft, J. E., Motwani, R., & Ullman, J. D. (2008). *Introduction to automata theory, languages, and computation*. Pearson.

References:

1. M. Sipser, *Introduction to the Theory of Computation*, Thomson Course Technology
2. E. Engeler, *Introduction to the Theory of Computation*, Academic Press.

Pokhara University
Faculty of Science and Technology

Course No.: ELE 270 (3 Credits)

Course title: Control System (3 -1- 2)

Nature of the course: Theory and Practical

Level: Undergraduate

Full marks: 100

Pass marks: 45

Time per period: 1 hour

Total Periods: 45

Program: BE Electrical and
Electronics Engineering

1. Course Description

The course aims to provide students with the necessary foundational knowledge about important control system parameters and their applications to engineering process.

2. General Objectives

The course is designed with the following general objectives:

- To become familiar with state space and transfer function representations while modelling linear-time-invariant systems.
- To learn various methods of analyzing the time response, frequency response, and stability of the systems.
- To develop basic feedback controllers.

3. Methods of Instruction

Lecture, tutorials, assignment, laboratories.

4. Course in Detail

Specific Objectives	Contents
To provide basic concept on control system	Unit I: Basic concept on control system (2 hrs.) 1.1 Introduction to control system 1.2 History of control system 1.3 Open loop and closed loop control system 1.4 Classification and examples of control system
- To introduce mathematical modeling of control system components - To introduce the state space models	Unit II: Mathematical modeling of physical system (7 hrs.) 2.1 Introduction to mathematical modeling 2.2 Modeling in state space 2.3 State-space representation of scalar differential equations 2.4 Linearization of nonlinear mathematical models 2.5 Mathematical modeling of mechanical and electrical systems

- To provide basic concepts on transfer functions model with graphical representation - To understand the analysis of the system in time domain along with steady state error	Unit-III: System transfer functions and responses (9 hrs.) 3.1 Combinations of components of physical system 3.2 Transfer function and impulse-response function 3.3 Block diagram algebra and system reduction 3.4 Signal flow graphs 3.5 Time response analysis of test signals 3.6 Time response analysis of first order and second order system 3.7 Transient response characteristics 3.8 Effect of feedback on steady state gain, bandwidth, error magnitude and system dynamics
To introduce frequency response analysis of the system	Unit-IV: Frequency Response Analysis (9 hrs.) 4.1 Closed loop frequency response 4.2 Performance specification in frequency domain 4.3 Frequency response of standard first order and second order system 4.4 Bode plots 4.5 Polar plots 4.6 Nyquist plots 4.7 Nichols' chart and its application
To analyses stability of physical system	Unit-V: Stability (9 hrs.) 5.1 Definitions of stability and stable systems 5.2 Complex plane interpretation of stability, root locations and stability 5.3 Routh-Hurwitz criterion for stability analysis 5.4 Application of Routh's stability criterion to control system analysis 5.5 Root locus 5.6 Rules for construction of root loci 5.7 Analysis and design using root locus concept 5.8 Nyquist stability criterion
- To construct compensators for linear system in frequency domain - To discuss the effect of PID controller in closed loop system	Unit-VI Performance Specifications and Design of Control System (9 hrs.) 6.1 Time domain specifications: steady state error, response rate, error criterion, hard and soft limits on responses 6.2 Damping ratio, log decrement 6.3 Frequency domain specifications: Band width, response amplitude ratio 6.4 Design of compensators using Bode plot and Root locus plot : Lead compensators , Lag compensators and Lag-lead compensators

5. List of Tutorials

The instructor will cover the numerical portion of the course in this tutorial section. The detail division of numerical portion are given below:

1. Unit II: Numerical related to mathematical modeling of electrical and mechanical systems. (2 hrs.)
2. Unit III: Numerical related to block diagram reduction and transfer functions, signal flow graphs, and time response of first order and second order system. (3 hrs.)
3. Unit IV: Numerical related to frequency response of first order and second order system, Bode plot, Polar plot and Nyquist plot. (4 hrs.)
4. Unit V: Numerical related to stability analysis using Routh-Hurwitz criterion, root locus and Nyquist criterion. (3 hrs.)
5. Unit VI: Numerical related to design of compensators using Bode plot and Root locus plot. (3hrs.)

6. Laboratory work

1. To study the open loop and closed loop mode of D.C. motor and familiarization with different components in D.C. motor control module.
2. To study effects of feedback gain and time constants for closed loop speed control system and position control system.
3. To study time domain response of first order and second order system.
4. Study of pole and zero plot and root locus plot.
5. To study frequency response of first order and second order system.
6. To study Bode plot and Nyquist plot.
7. Design of compensators using Bode plot and Root locus plot.

8. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20
		Attendance & Class Participation	10%	
		Laboratory Report	10%	
		Viva	20%	
		Exam	60%	

	Total Internal	50
Full Marks: 50 + 50 = 100		

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

9. Prescribed Books and References

Text Books

1. K. Ogata, *Modern Control Engineering*, 5th Edition, PHI, 2012.
2. B.C. Kuo, *Automatic Control Systems*, 7th Edition, PHI of India, 1995.
3. M. Gopal, *Control System- Principles and Design*, 5th Edition, Tata McGraw Hill, 2012.

References

1. S.N. Nise, *Control System and Engineering*, John Wiley and Sons, 2011.
2. J. Nagrath and M. Gopal, *Control System Engineering*, 5th Edition, New Age International Publishers, 2007.
3. S.K. Bhattacharya, *Control System Engineering*, 3rd Edition, Pearson, 2013.

Pokhara University
Faculty of Science and Technology

Course No.: ELE 272 (2 Credits)

Course title: Electrical Machine II (2-1-2)

Nature of the course: Theory & Practical

Level: Undergraduate

Full marks: 100

Pass marks: 45

Time per period: 1 hour

Total Periods: 45

Program: BE Electrical and
Electronics

1. Course Description

The purpose of the course is to provide the fundamental concept of three phase alternator, parallel operation and Synchronous motor. Furthermore, this course provides the idea of operating principle and characteristics of emerging machines.

2. General Objectives

The course is designed with the following general objectives:

- To provide the concept of fundamental of electromagnetic energy conversion
- To provide the concept of three phase alternator, parallel operation and Synchronous motor.
- To familiarize the students with construction, working principle, and characteristics of different types of emerging machines

3. Methods of Instruction

Lecture, tutorials, assignments, laboratories

4. Course in Detail

Specific Objectives	Contents
• Describe the construction, working principle and characteristics of three Phase AC generator • Discuss the modeling of three phase AC generator and its transient behavior and capability curves • Explain and analyze parallel operation of three phase alternator	Unit I: Synchronous Generator (14 hrs.) 1.1 Constructional details, Operating principle, EMF equations, 1.2 Winding factors, Harmonics in generated voltages and their suppression 1.3 Armature reaction, synchronous impedance, phasor, diagram 1.4 No Load and Short Circuit Test 1.5 Voltage Regulation and Efficiency 1.6 Two reaction Theory for salient pole machine, Vector diagram with d-q axis and Power Equation 1.7 Transient condition of an alternator, Transient and sub-transient reactance 1.8 Synchronous Generator Rating and P-Q Capability Curves 1.9 Parallel Operation : Need of Parallel Operation ,

	Condition requirement , Synchronization and Methods for Synchronization 1.10 Real and reactive power sharing 1.11 Operation of Generators in Parallel with Large Power System, Concept of infinite bus 1.12 Synchronizing with infinite bus bars, synchronizing power
- Describe the construction, working principle of Synchronous Motor - Discuss and analyze the effect of load and excitation	Unit II: Synchronous Motor (6 hrs) 2.1 Theory of operation, Equivalent Circuit and Characteristics 2.2 Vector diagram with d-q axis and Torque Equation 2.3 Steady-State Synchronous Motor Operation 2.3.1 Effect of Load and Excitation 2.3.2 V Curves 2.3.3 Synchronous Condenser 2.4 Starting Synchronous Motors: Methods and Performance 2.5 Damper Winding and Hunting effect 2.6 The Effect of Damper windings on Motor Stability .
- Explain the basic concept of electromechanical conversion and torque production of machines - Explain the construction , operation and performance of Emerging Machines	Unit III: Electromechanical conversion and Special Machines (10 hrs.) 3.1 Field and co energy concept, Magnetic Force and Torque from Energy and Co-energy 3.2 Single and doubly excited magnetic field systems, Reluctance torque in rotating machine 3.3 Torque Equation: Cylindrical rotor machine, Salient pole machine and Permanent Magnet Machine 3.4 Permanent Magnet Synchronous motor: 3.4.1 Permanent Magnets Materials, B-H loop and demagnetization characteristics 3.4.2 Construction Principle of operation, Emf equation 3.4.3 Phasor Diagram, Torque equation of BLPM sine wave motor and Permissible torque-speed characteristics 3.5 Permanent Magnet Brush Less DC Motor: 3.5.1 Constructional feature, Principle of operation of brushless pm dc motor and Emf and Torque 3.5.2 Torque- speed characteristics and Commutation with 120° and 180° magnet arc 3.6 Synchronous Reluctance Motor: 3.6.1 Construction and Operation 3.6.2 Torque – speed characteristics and Phasor diagram

	3.7 Stepper Motor: 3.7.1 Classification, Construction and Principle of operation 3.7.2 Characteristics of stepper motor: Static characteristics. Dynamic characteristics and Torque-speed characteristics 3.7.3 Micro stepping control of stepping motor 3.8 Switched Reluctance Motor: 3.8.1 Construction and operation 3.8.2 Torque-speed characteristics and Torque Speed Capability Curve
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8 Laboratory Work

1. Determination of alternator performance conducting short circuit and open circuit test in Laboratory (Synchronous Impedance Method).
2. Determination of alternator performance conducting short circuit and open circuit test in Laboratory (Zero Pf Method).
3. Determination of V curve of Synchronous Motor
4. Synchronization of three phase alternator with supply
5. Determination of power sharing of parallel connected alternators and study of effects of excitation and input power
6. Simulation of Emerging Machines Performance in suitable software

9 List of Tutorials

The instructor will cover the numerical portion of the course in this tutorial section. The detail division of numerical portion are given below:

1. Unit I: Numerical related to alternator generated emf. test and Performance and parallel operation. (7 hrs.)
2. Unit II: Numerical related to synchronous motor performance, effect of excitation and load and Synchronous Condenser (4 hrs.)
3. Unit III: Numerical related to Electromechanical conversion and Special Machines. (4 hrs.)

10 Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	

		Practical		20
		Attendance & Class Participation	10%	
		Project Report	10%	
		Viva	20%	
		Exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

11 Prescribed Books and References

Reference

1. E. Fitzgerald, C. Kinsley & S. Dumans, *Electric Machinery*, Tata McGraw Hill, India Ltd. 1984
2. Langsdorf, Alexander S. *Theory Alternating Current Machine*, McGraw-Hill book company, inc. First Edition 1937
3. Puchstein, A. F., Lloyd, T. C. *Alternating Current Machines*, New York John Wiley & Sons, Inc. London: Chapman and Hall, Limited 1942
4. *Brushless permanent magnet machine and Switch reluctance motor* by T.J.Miller
5. Krishana R. *Permanent Magnet Synchronous and Brushless DC Motor Drives*, Taylor & Francis Inc. First Edition 2017 ISBN: 9780824753849, 9780824753849
6. *Theory and Performance of Electrical Machine* by J.B.Gupta
7. Bhag S. Guru and Huseyin R.Hiziroglu *Electrical Machinery and Transformers*, New York Oxford University Press 2001 Third Edition

Pokhara University
Faculty of Science and Technology

Course No.: ELE 274 (3 Credits)

Course title: Transmission and Distribution System (3 -1- 0)

Nature of the course: Theory

Level: Bachelor

Full marks: 100

Pass marks: 45

Time per period: 1 hour

Total Periods: 45

Program: BE Electrical and
Electronics Engineering

1. Course Description

The purpose of the course is to give the students requisite basic knowledge about key parameters of transmission and distribution of a modern power system. Furthermore, this course provides the idea of modelling and analyzing the performance of transmission and distribution systems.

2. General Objectives

The course is designed with the following general objectives:

- To impart the knowledge on the structure of electric power systems and develop equivalent circuit models.
- To develop the ability to carry performance evaluation related computations of transmission and distribution systems.

3. Methods of Instruction

Lecture, tutorials, assignments, etc.

4. Course in Detail

Specific Objectives	Contents
• To introduce the structure and components of power system • To impart knowledge for necessity of different voltage levels in power system with appropriate reasoning • To distinguish transmission and distribution systems	Unit I: Introduction to Power System (5 hrs.) 1.1 Basic concept of electricity generation including their sources and consumption of electricity 1.2 Power delivery system, interconnection of different generations and loads 1.3 Single phase and Polyphase transmission lines, single circuit and double circuit lines, advantages of higher voltage transmission 1.4 Components of power system; substations, transmission lines and distribution system 1.5 Concept of HVDC transmission
• To familiarize students with various physical components used in overhead lines and underground cables	Unit II: Overhead lines and underground cables (6 hrs.) 2.1 A brief introduction to types of supporting

To compute potential distribution over a string insulators and string efficiency	structures, tower footing grounding and line conductors 2.2 Insulators for overhead lines: types of insulators, their constructions and performance, material used- porcelain, toughened glass and polymer (composite) 2.3 Potential distribution over a string of suspension insulators. String efficiency, Methods of increasing string efficiency. 2.4 Arcing horns, Line vibration and vibration dampers, bird diverters, bird flappers, etc. 2.5 Overhead line protection against lightning; ground wires 2.6 Underground Cable: Types of cables, constructional features, conducting, and insulating materials used in cables, insulation resistance, thermal rating.
- To impart knowledge about key electrical parameters of transmission and distribution lines - To develop expressions for the computations of transmission line parameters - Computing underground cable related parameters and comparison with overhead lines	Unit-III: Line Parameters computations (12 hrs.) 3.1 Introduction to line parameters: resistance, inductance and capacitance. 3.2 Fundamental principle of line inductance and capacitance computations and expressions 3.3 Skin and proximity effects 3.4 Inductance and capacitance of composite conductors, geometric mean radius (GMR) and geometric mean distance (GMD). 3.5 Calculation of inductance and capacitance of single phase and three phase lines with equilateral spacing, unsymmetrical spacing, double/multi circuit and transposed lines. 3.6 Introduction to bundled conductors, inductance, and capacitance computation for bundled conductors. 3.7 Earth effect of overhead line capacitance 3.8 Inductance and capacitance computation for cables 3.9 Grading of cables: capacitance and inter-sheath 3.10 Cable charging current, dielectric loss 3.11 Comparison between AC and DC cables, limitations of cables
- To obtain the equivalent circuits for the transmission lines based on distance - To evaluate the voltage regulation and efficiency of transmission line - To compute the performance	Unit-IV: Performance of Transmission Lines (14 hrs.) 4.1 Per unit computations and their advantage in power system 4.2 Single phase equivalence and single line diagram of a 3-phase transformers and lines

indicators for a underground cables	4.3 Classification of lines: short, medium, and long transmission line 4.4 Nominal T and pi equivalent circuits and distributed line model of transmission line 4.5 ABCD parameter computations for short and medium transmission line 4.6 Long transmission line, voltage waves, surge impedance and propagation constants. 4.7 ABCD parameter computations for long transmission line, equivalent pi model 4.8 Power flow through transmission lines, receiving and sending end power circle diagram 4.9 Voltage regulation and efficiency computations of transmission lines 4.10 Ferranti effects on long transmission lines, concept of surge impedance loading, countermeasures for Ferranti effects 4.11 Shunt compensation: performance evaluation with shunt inductive and capacitive compensation
• To acquaint with distribution system terminologies and understand the peculiarities of primary and secondary distribution system • To familiarize with consumers demand characteristics • to compute voltage drop and efficiency of a distribution line	Unit-V: Distribution system (8 hrs.) 5.1 Distribution system voltage level and terminologies: substation, primary and secondary distribution, distribution transformers, load centers 5.2 Brief introduction about Nepalese distribution system 5.3 Primary AC distribution systems: Radial feeders, parallel feeders, loop feeders and interconnected network system. 5.4 Secondary AC distribution systems: Three phase 4 wire system and single phase 2 wire distribution, AC distributors with concentrated loads 5.5 Consumer demand characteristics, load curve and load duration curve, load factor, coincidence factor. 5.6 Power loss and voltage drop computation in AC distribution lines

5. List of Tutorials

The instructor will cover the numerical portion of the course in this tutorial section. The detail division of numerical portion are given below:

1. Unit II: Numerical related to potential distribution over a string of suspension insulators and string efficiency. (1 hrs.)

- Unit III: Numerical related to calculation of inductance and capacitance of overhead transmission lines and underground cables, grading of cables & charging current and dielectric loss of cables. (6 hrs.)
- Unit IV: Numerical related to Per unit computations, ABCD parameters of short medium and long lines, voltage regulation and efficiency, Ferranti effects and surge impedance loading. (6 hrs.)
- Unit V: Numerical related to consumer load characteristics and distribution system voltage drop and power loss computations. (2 hrs.)

6. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20
		Attendance & Class Participation	10%	
		Project Report	10%	
		Viva	20%	
		Exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Books

- William Stevenson, Jr., John Grainger, Power System Analysis, McGraw-Hill Education
- M.L. Sony, P.V. Gupta et al., A textbook on Power System Engineering, Dhanpat rai & Co
- C. L. Wadhawa, Electrical Power System, New Age International Publication

References

1. Kirtley, James, Electric Power Principles: Sources, Conversion, Distribution and Use, Wiley
2. O.E. Elgerd, Electric Energy Systems Theory, TMH Publishing Company

Pokhara University
Faculty of Science and Technology

Course No.: ELE 276 (3 Credits)

Course title: Electrical Machine and Control (3-1-2)

Nature of the course: Theory & Practical

Level: Undergraduate

Full marks: 100

Pass marks: 45

Time per period: 1 hour

Total Periods: 45

Program: BE Electronics and
Communication

1. Course Description

The purpose of the course is to provide the fundamental concept DC and AC industrial motor principle of operation and terminal characteristics motor Also, this course provides the idea of basic power switching devices operation and converters and characteristics. Furthermore the courses provides phase control of industrial motor and sequential control of Industrial Drives system

2. General Objectives

The course is designed with the following general objectives:

- To provide the concept of DC and AC industrial motor construction, operation and characteristics with suitable model and equations
- To provide the in-depth knowledge of power semiconductor switches and different converter
- To familiarize the students with speed and torque control of DC and AC motor drives

Methods of Instruction

Lecture, tutorials, assignments, laboratories

3. Course in Detail

Specific Objectives	Contents
• Able to describe the operation of solid state switches, their turn on and turn off delay, switching losses and method for turn on and turn off	Unit I Power Semi-Conductor Switches (5 hrs.) 1.1 Ideal and Generic-switch Switching Characteristics and Losses 1.2 Power Diode: Structure, operation, static and Dynamic Characteristics, reverse recovery time and types 1.3 Power Transistor: Structure, operation, Switching Characteristics 1.4 Thyristor: Structure, turn on and Turn off Processes, Operation of TRIAC and DIAC, Protection and Snubber circuit 1.5 Metal Oxide Semiconductor Field Effect Transistor: Structure, operation as switch, switching characteristics

	1.6 Insulated Gate Bipolar Transistor (IGBT): Structure, operation as switch and switching characteristics 1.7 Gate Turn off Thyristor (GTO): Operation mechanism and Switching Characteristics 1.8 Other Emerging switching devices
Explain the basic concept of DC Motor and its speed control mechanism	Unit II: Direct Current Machines (4 hrs.) 2.1 DC Motor: Construction, operation principle Back emf, Torque production, Types and Equivalent Circuit 2.2 Terminal Characteristic: Separately excited, Series and Shunt Motor 2.3 Speed Control: Separately excited, Series and Shunt Motor 2.4 Open loop model for the DC separately excited and shunt motor 2.5 Transfer function for the open loop model for voltage and torque disturbances
Explain the basic concept of Induction Motor, Permanent Magnet Motor and their speed control mechanism	Unit III: Three Phase Induction Motor (4 hrs) 3.1 Construction, Types, Principle of operation and concept of slip 3.2 Equivalent Circuit, Power Flow, Torque Equation and Torque-Speed Characteristics 3.3 Control of Motor Characteristics: Cage Rotor Deep-Bar and Double-Cage Rotor Designs 3.4 Speed Control of Induction Motors: Pole Changing, Line Frequency, Line Voltage, Rotor Resistance and VVVF methods 3.5 Permanent Magnet synchronous Motor: Operation and Speed Cotel
Able to analyze the performance of AC to DC controlled and Un-controlled Rectifier	Unit IV: AC to DC Converter (6 hrs.) 3.1 Uncontrolled Rectifier: 3.1.1 Single Phase Half -Wave Rectifier and full Wave rectifier: Circuit Diagram, Operation principle, and performance parameters 3.1.2 Three Phase Half and full -Wave Rectifier: Circuit Diagram, Operation principle, and performance parameters 3.2 Controlled Rectifier: 3.2.1 Single Phase Semi and full Converter: Circuit Diagram, Operation principle, and performance parameters 3.2.2 Three Phase Semi and full Converter: Circuit Diagram, Operation principle, and performance parameters

	4.3. Effect of Source inductance 4.5 Types of control for harmonic distortion improvement
Explain the operation of inverter and their performances	Unit V: Inverter (4 Hrs.) 5.1 Basic Topology: Half Bridge inverter with R and RL load, Full Bridge inverter with R and RL load, 5.2 Three Phase Inverter 5.3 Performance of single and Three Phase Inverter 5.4 PWM inverter: Sinusoidal Pulse Width Modulation, Unipolar and Bipolar Switching and Performance of SPWM inverter
Describe the operation of DC-to-DC converter and their performances	Unit VI DC-DC converters (4 hrs) 6.1 Step Down converters: Principle of operation, Buck converter operation: voltage ripple and current ripple and its control 6.2 Step up converters: Principle of operation, Boost converter operation: voltage ripple and current ripple and its control 6.3 Four quadrant operation
- Able to Discuss the performance of basic open loop and closed loop control systems - Demonstrate transfer function model for electrical and mechanical systems - Analyze the stability of a system based on time response - Analyze the stability of control systems through time domain and frequency domain approach	VII Feedback and Control Principles (8 hrs) 7.1 Introduction to Control Systems: Introduction, Dynamic Equation of Physical System, Block diagram. Transfer function. Examples of Control Systems and Closed-Loop Control versus Open-Loop Control 7.2 Transient Response and Steady-State Response: Transfer Function analysis of first order system with unit step and unit ramp impulse, Transfer function analysis of second order system with unit step impulse and Definitions of Transient-Response Specifications 7.3 Routh's Stability Criterion and Application of Routh's Stability Criterion to Control System Analysis 7.4 Effects of Proportional, Integral and Derivative Control Actions on System Performance, Steady-State Errors in Unity-Feedback Control Systems Example Problems and Solution 7.5 Frequency-Response Analysis: Frequency-Response, Bode Plots, Gain margin, Phase margin system stability, transient characteristics: damping ratio, percent overshoot bandwidth, response speed, steady-state error – Gain adjustment in the frequency domain

- Perform speed control of a given DC motor using controlled rectifier - Perform speed control of a given DC motor using DC to DC converter - Able to design and analysis closed loop controller for DC drives	Unit VIII: Converter Fed DC drives (6 hrs) 8.1 Open loop operation of AC to DC controlled rectifier fed DC Motor Drives, Continuous and Discontinuous operation and Regenerative Braking 8.2 Open loop operation of DC-to-DC converter fed DC Motor Drives, Continuous and Discontinuous operation 8.3 Closed Loop Control: Introduction, Controller Objectives, Cascade control Structure, controller design: Torque control loop, Speed control Loop and Position control Loop 8.4 Example of controller Design
- Illustrate solid state speed control of a 3-phase induction motor by voltage control and frequency control method. - Explain solid state speed control of a 3-phase induction motor by variable frequency drive and rotor resistance control method	Unit IX: AC drives (5 hrs) 9.1 Speed control of three phase induction motor: Stator Voltage control, voltage/frequency control, Rotor Chopper 9.2 Control with AC voltage regulators-applications 9.2 Slip power recovery scheme 9.3 Closed Loop control;

8 Laboratory Work

- Speed Control of DC Separately excited DC motor
- Characteristics of D.C. and AC. Servomotors
- Frequency response of second order system.
- Step response of second order system
- D.C. and AC Position control system
- Performance of P, PI and PID Controller on system response
- Simulation of converter, chopper and Inverter
- Simulation of DC separately excited DC motor for the voltage and Torque disturbance in open loop
- Simulation of DC separately excited DC motor for the voltage and Torque disturbance in closed Loop
- Simulation for Speed control of three Phase Induction Motor

9 List of Tutorials

The instructor will cover the numerical portion of the course in this tutorial section. The detail division of numerical portion are given below:

- Unit I: Numerical related to Calculation of switching Losses. (1 hrs.)
- Unit II: Numerical related to Dc Machine Speed Control (1 hrs.)
- Unit III: Numerical related to Induction machine speed control. (1 hrs.)
- Unit IV: Numerical related to AC to DC uncontrolled and Controlled rectifier performance (2 hrs.)
- Unit V: Numerical related to single and three Phase inverter. (1 hrs)
- Unit VI: Numerical related to DC chopper. (1 hrs)

7. Unit VII: Numerical related to basic control system: Mathematical modeling, Transfer function. response of 1st order and second order system, Routh stability effect of P, PI and PID on system response and Bode Plot and stability . (5 hrs.)
8. Unit VIII: Cascade controller design for separately Excited DC motor Drive . (3 hrs)

10 Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20
		Attendance & Class Participation	10%	
		Project Report	10%	
		Viva	20%	
		Exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

11 Prescribed Books and References

Reference

1. Ned Mohan, Undeland and Robbin, „Power Electronics: converters, Application and design“, John Wiley and sons. Inc, Newyork, 2006.
- 2.. Rashid M.H.,, Power Electronics-Circuits, Devices and Applications,, Prentice Hall India, New Delhi, 2009. 3. P.C Sen., „Modern Power Electronics“, Wheeler publishing Company, 1st Edition, New Delhi, 2005.
3. P.C Sen, „Thyristor DC Drives“, John Wiley and Sons, New York, 1991. 2. R. Krishnan, „Electric Motor Drives – Modeling, Analysis and Control“, Prentice-Hall of India Pvt. Ltd., New

4. Ogata K., Modern Control Engineering, Prentice-Hall of India Pvt Ltd., New Delhi, 3rd edition, 2000.
5. E. Fitzgerald, C. Kinsley & S. Dumas, *Electric Machinery*, Tata McGraw Hill, India Ltd. 1984
6. Bhag S. Guru and Huseyin R.Hiziroglu Electrical Machinery and Transformers, New York Oxford University Press 2001 Third Edition
7. Bimal K. Bose, „Modern Power Electronics & AC Drives“ Phi Learning Pvt. Ltd, 2nd Edition, New Delhi, 2002
8. Fundamental of Electrical Drives, G.K. Dubey, New Age International Publication.